

Gaussian Quantum Steering and Entanglement Dynamics in Coupled Lossy Waveguides



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Dedication

I dedicate this work to my parents, whose unconditional support and encouragement have been the backbone of my academic journey. I also dedicate it to my husband, whose confidence in me and constant support have greatly encouraged and strengthened my academic journey.

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I would like to express my sincere gratitude to my supervisor, Dr. Shakir Ullah, for his guidance, valuable suggestions, and constant support throughout this research. His knowledge in quantum optics and quantum information has helped me learn and think more deeply, and I am truly thankful for that. I am grateful to the Department of Physics, Quaid-i-Azam University, for providing the research environment and facilities needed for this work. A special thanks to my husband, whose patience, support, and motivation kept me going. His belief in me made this work possible. Lastly, I acknowledge the efforts of researchers whose previous studies in quantum steering and laser physics laid the groundwork on which this research is built

Abstract

List of Abbreviations

CV	Continuous Variable
DV	Discrete Variable
1sDI	One-Sided Device-Independent
QKD	Quantum Key Distribution
1sDI-QKD	One-Sided Device-Independent Quantum Key Distribution
PPT	Positive Partial Transpose
TMGS	Two-Mode Gaussian State
NOON	N-Photon State
HOM	Hong-Ou-Mandel (interference)
QIT	Quantum Information Theory
EPR	Einstein-Podolsky-Rosen
EN	Entanglement via Logarithmic Negativity
E_N	Logarithmic Negativity Measure
$S^{A \rightarrow B}$	Steering from Party A to Party B
V	Covariance Matrix
ρ	Density Matrix
γ	Loss Rate (decay)
J	Coupling Strength (waveguides)
γ/J	Loss-to-Coupling Ratio

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Chapter 1

Introduction

1.1 Background and Motivation

Quantum correlations is one of the basic aspect of quantum mechanics required for many current technologies in quantum information [1, 12]. The field of quantum information started with the aim of controlling individual quantum bits (qubits), and has led to important concepts, such as quantum computing [1], quantum cryptography [2], and quantum teleportations [3]. Later researchers discovered an immensely powerful alternative by using continuous-variable (CV) quantum information carriers instead of qubits, offering the quantization of degrees of freedom of bosonic systems, such as the electromagnetic field [11, 4]. These systems are crucial in quantum communication, as well as for very advanced methods, that are limited only by quantum physics, and are used to detect, sense, and image [11].

In quantum processes involving CVs, it is frequently of interest to consider the Gaussian realm, i.e., Gaussian states and Gaussian transformations, and this is a space where the mathematics is simple and well-organized. Further, the required optical devices in the laboratory, (such as beam splitters, phase shifters, and squeezers) are simple to access and are readily available in large quantities [11]. This area is referred to as Gaussian quantum information processing, and allows a vast range of important tasks to be performed. The most important of these protocols are quantum correlations, most notably entanglement and quantum steering, which are key to almost all quantum information protocols being implemented successfully [11].

In 1935 the idea of quantum entanglement gained much attention with the Einstein-Podolsky-Rosen (EPR) paradox raising the question of whether quantum mechanics provides a complete description of reality [7]. Later that year, Schrodinger referred to this effect as entanglement and also proposed a new form of quantum correlation which is now called quantum steering [8, 19]. These discoveries formed the basis of the modern theory of quantum correlation, which are crucial for various quantum information protocols [25, 1].

The phenomenon known as quantum steering was introduced in 1935 [8]. It is a form of quantum correlation that has received considerable attention in quantum information theory, especially after its formalisation in 2007 [38]. Schrodinger first described this phenomenon as "magic" as the effect does not facilitate transmission of information, and it is the ability of one party (Alice) to measure or steer the wave function of another distant party [19]. One common notion in the modern theory is the inability of the

conditional quantum states of one party (Bob) to be described by a local hidden state (LHS) model. A joint bipartite quantum state ϱ_{AB} is said to be steerable if Bob is forced to believe that Alice is influencing his system through some form of action at a distance. conversly,when (LHS) exists for the shared state,then such a bipartite system is referred to as an unsteerable state [19].

One-sided device-independent quantum key distribution (QKD) is essential due to quantum steering [41], in which one party is confident in their measurement equipment, whereas the other is not, which is clear in the possibility of secret key distillation and demonstrates the state steerability [41, 19]. It also has randomness certification in this asymmetric case with more extractable randomness than the Bell case with full device-independence [42]. Besides, steering offers an operational benefit in subchannel discrimination tasks and is conceptually equivalent to quantum teleportation and analysis of security in secret sharing protocols [44, 19].

Quantum entanglement is recognized as a fundamental essence of quantum mechanics and quantum information theory [25]. Entanglement is described as a physical phenomenon, in which the associated particles remain correlated irrespective of their physical distance between them [7]. This property is deemed important for the practical implementation of the quantum information processing systems. Its importance can be seen in numerous important technological domains such as, quantum computing, quantum cryptography, and quantum teleportation playing a significant part [1, 25]. Particularly, entanglement is viewed as being crucial to illustrate the supremacy of a quantum computer when comparing it to its classical analogs [45]. In connection to this, in coupled lossy waveguides, entanglement is an important concept, as these waveguides can be used as basic parts of quantum circuits [46, 47, 51]. Although the real devices always have some losses, the devices are still useful to quantum information tasks and it is therefore necessary to know how quantum entanglement and quantum steering really behaves in them [56, 28, 25].

An optical waveguide is a structure that guides light along a desired path. Coupled waveguides are very important because they act as basic building blocks for quantum circuits. These waveguides allow efficient control of light and can be effectively used to manipulate light to accomplish quantum information processing, where the manipulation of individual photons in a waveguide is fundamentally involved [46, 50, 47]. In addition, coupled silica waveguides have been extended to make multimode integrated interferometers which can produce arbitrary quantum circuits, as well as can produce

two and four photon entangled states analogous to NOON states [51, 46, 52]. In addition to computation, such structures can be considered to study other important quantum effects such as quantum interference, entanglement and quantum walk [47, 53, 50]. An important point of these structures is that it is dealing with real material loss and it still finds that the waveguides hold up well. They are able to maintain their entanglement even though the loss is quite strong [28]. This shows that coupled lossy waveguides can still work as useful parts of quantum circuits, making them suitable for many quantum information applications [28, 11].

As quantum technologies have advanced, optical waveguide-based photonic platforms have become the focus of modern quantum hardware. In these Waveguide based devices light is trapped in small, stable cores with high index, and small and extremely scalable quantum circuits are constructed [46, 50]. By making two waveguides close to each other, all the evanescent fields overlaps, leading to evanescent coupling mechanism that enables controlled energy exchange just like a beam splitter. This interaction is the foundation on which to apply on-chip quantum gates and other necessary photonic operations are applied [28, 46, 50].

however, in real devices, some amount of optical loss and noise from the environment is always present [56]. The energy of the photons decreases as they move, leading to an effective interaction with the empty vacuum modes; the interaction tends to destroy the quantum state of a system, and destroys the distinct quantum correlations between the modes [54, 56]. Such perturbations destroy quantum correlations, i.e quantum entanglement and quantum steering [25, 19], and they pose an enormous limitation to photonic architectures which require either large propagation ranges or very precise interactions between gates [28, 11, 17].

For this reason, understanding how quantum correlations behave in coupled waveguides when they experience losses is essential for determining whether integrated quantum photonic systems can work in practice. By studying how entanglement and steering affects under an attenuation, we can evaluate how robust these quantum resources are and assess the viability of waveguide-based circuits for future quantum technologies.

This research analyzes the time-dependent dynamics of quantum steering and entanglement in a passive two-mode directional coupler with realistic loss, using the Gaussian-states and Gaussian transformation formalism. The study examines how several key parameters—including the loss coefficient κ , the initial nonclassical depth τ , the purity

μ , coupling parameter J and the input squeezing r of the Gaussian state—collectively determine the preservation or degradation of quantum steering and entanglement as the fields propagate through lossy waveguides.

1.2 Thesis Layout

This thesis is structured as follows:

Chapter 2 begins with an overview of quantum states of light discussing CV quantum states particularly Gaussian states together with their representation through quadrature operators, wigner functions and covariance matrices. After that the discussion moves towards optical waveguides which highlights evanescent coupling, loss and decoherence in waveguides and applications of optical waveguides in quantum photonics. Moreover, the chapter then reviews the two important quantum correlations, i.e., quantum steering and quantum entanglement with their quantification, significance, and applications. Last but not the least, the applications and relevance of the optical waveguides in quantum circuits and photonics, and the applications of quantum steering and quantum entanglement are overviewed. After that Chapter 3 starts with an overview of our physical system. It then describes the system's hamiltonian. It is followed by the discussion of the physical interpretation of the coupling. The chapter then discusses the open system dynamics and loss modelling which contains the quantum liouville equation and lindblad operator for waveguide loss. It then describes equations of motion for our system, the expectation value formalism and the moment equations and their solutions. With these moment equations in hand, we calculated covariance matrix and then quantified quantum steering and quantum entanglement. The quantum hierarchy relationship between entanglement and steering is also explained which is then proved from our results. A special type of Gaussian states called squeezed states, its covariance matrix which we used to evaluate entanglement and steering is discussed at the end of this chapter. Chapter 4 is all about the results, begins the discussion about the dynamical behaviour of Gaussian quantum steering and entanglement in a coupled lossy waveguide system. First of all it took general case in which all the parameters κ , τ , μ , are kept constant. It then takes specific cases one by one to see the effect of τ , μ and κ where $J=1$ for all. This chapter then explains the results for the special type of Gaussian states such as squeezed states. Initially, we consider describes ideal non lossy case and then discusses the effect of losses on quantum steering and entanglement in squeezed states. Lastly, the effect of coupling

parameter J on quantum steering and quantum entanglement is explained. The chapter ends with the summary of the results and discussion. In Chapter 5 we have written the conclusion, where we discussed the full summary of the thesis research work and our key findings. We have added all the bibliography/citations in Chapter 6.

Chapter 2

Literature Review:

This chapter gives a review of the theoretical background of quantum correlations in integrated photonic circuits. It begins with a review of continuous-variable quantum states, particularly of Gaussian states, the properties of which are fully characterized by a covariance matrix, extracted out of the field quadratures. This is in turn followed by a discussion of the physical platform of optical waveguides, how the light is localized within the guiding medium and interaction between adjacent modes through evanescent interaction, provide ordinary optical attenuation (α) couples the system to the vacuum and causes the system to decay (Γ), making it hard to preserve quantum information. The chapter also summarizes two important quantum resources discussed in this work, quantum steering, denoted by the Gaussian steering measure, denoted by the symbol (S), and quantum entanglement, characterized by the logarithmic negativity E_N . All the above factors together are the conceptual framework that must be used to research the dynamical development of these resources in coupled waveguide architectures. This chapter ends with the discussion of the vital uses of entanglement, steering, and waveg-

udes.

2.1 Quantum States Of Light

The quantum states of light are defined to be quantum-mechanical characterisations of the electromagnetic field, in which the field is considered a quantised bosonic system under a quantum system instead of a classical wave, which are characterised by state vectors of density operators acting on the Hilbert space of modes of a field [55]. These states, unlike classical light, can have nonclassical characteristics, including squeezing, entanglement and quantum correlations, which are of paramount importance to Continuous-variable quantum information and quantum technologies [11]. Quantum states of light are classified into pure and mixed states, Fock (number) states, coherent states, squeezed states, entangled states and more generally the Gaussian and non-Gaussian states [55, ?]. In terms of information encoding and measurement information quantum states of light can either be modeled in a discrete-variable model, in terms of photon-number observables, or in a Continuous-variable model, where information is encoded in field quadratures with continuous spectra [4, 11].

2.1.1 Continious Variable States

In quantum mechanics, a Continious variable state is a quantum state where the physical observables i.e, momentum and position vary continiously unlike discrete variable state where observables can take only discrete values. These system are illustrated with in an infinite-dimensional Hilbert space [4]. In these system the principal variables are the quadrature operators which obey specific commutation relations that form the foundation of the system's quantum structure [10]. Continious variable quantum states are classified into two categories: Gaussian and non-Gaussian states [12].

2.1.2 Guassian States

In a Continious variable system quantum system, a Gaussian state is a special kind of quantum state whose fourier representation has a guassian form in phase space. These states are completely described by their first and second moments [11]. The quadrature operators are [55]

$$\hat{Q} = \frac{1}{\sqrt{2}}(\hat{a} + \hat{a}^\dagger), \quad \hat{R} = \frac{i}{\sqrt{2}}(\hat{a}^\dagger - \hat{a}), \quad (2.1)$$

which satisfy the canonical commutation relation

$$[\hat{Q}, \hat{R}] = i. \quad (2.2)$$

The Wigner function of a Gaussian state is

$$W(\hat{\mathbf{z}}) = \frac{1}{(2\pi)^m \sqrt{\det \boldsymbol{\sigma}}} \exp \left[-\frac{1}{2} (\hat{\mathbf{z}} - \bar{\mathbf{z}})^T \boldsymbol{\sigma}^{-1} (\hat{\mathbf{z}} - \bar{\mathbf{z}}) \right]. \quad (2.3)$$

Here,

$$\hat{\mathbf{z}} = (\hat{Q}_1, \hat{R}_1, \dots, \hat{Q}_m, \hat{R}_m)^T$$

is the phase-space vector, whose components are denoted by $\hat{z}_i$,

$$\bar{\mathbf{z}} = \langle \hat{\mathbf{z}} \rangle$$

is the displacement vector, and

$$\sigma_{ij} = \frac{1}{2} \langle \hat{z}_i \hat{z}_j + \hat{z}_j \hat{z}_i \rangle - \langle \hat{z}_i \rangle \langle \hat{z}_j \rangle$$

is the covariance matrix.

All physical properties of the Gaussian states such as purity, entropy, steering and entanglement can be determined directly from the covariance matrix without full density operator because of the gaussian form of Wigner function. typical examples of gaussian states include the coherent states, squeezed states, thermal states and vacuum state [11, 12].

2.1.3 Squeezed States

A special type of Gaussian state are squeezed states. As a Gaussian states, they are fully determined by their first and second statistic moments [11, 12]. These initial moments are associated with the averages of the quadrature operators which are displacements of the phase space [11]. The second moments are grouped into a positive-definite, symmetric and real matrix called the covariance matrix [11, 12]. The action of a displacement

and a squeezing on the vacuum state could create any pure Gaussian state [12].

The theoretical description of squeezed states begins with the bosonic annihilation and creation operators $\hat{a}$ and $\hat{a}^\dagger$, which satisfy the canonical commutation relation

$$[\hat{a}, \hat{a}^\dagger] = 1.$$

The field quadrature operators, analogous to position and momentum, are defined as

$$\hat{Q} = \frac{1}{\sqrt{2}} (\hat{a} + \hat{a}^\dagger), \quad \hat{R} = \frac{i}{\sqrt{2}} (\hat{a}^\dagger - \hat{a}).$$

[11, 12] These quadratures obey the canonical commutation relation [55]

$$[\hat{Q}, \hat{R}] = i,$$

which leads to the uncertainty relation

$$\Delta Q \Delta R \geq \frac{1}{2}.$$

[39, 12]

A single-mode squeezed vacuum state is generated by applying the unitary squeezing operator $\hat{S}(\zeta)$ to the vacuum state $|0\rangle$:

$$\hat{S}(\zeta) = \exp\left[\frac{1}{2} (\zeta^* \hat{a}^2 - \zeta \hat{a}^{\dagger 2})\right]. \quad (2.4)$$

Here, the complex squeezing parameter $\zeta = s e^{i\theta}$ determines the squeezing strength s and the squeezing phase θ [39, 12].

The Heisenberg picture represents the squeezing transformation that actually elongates and rotates the quadrature axes. In the case of rotated quadratures that are duly rotated Y_1 and Y_2 , the variances become

$$V(Y_1) = \frac{1}{2} e^{-2s}, \quad V(Y_2) = \frac{1}{2} e^{2s},$$

It is evidently indicating that quantum noise in one quadrature has been reduced below the amount in the vacuum at the cost of increasing the fluctuations in the conjugate quadrature [39].

One of the most important results that were reported in the early papers on quantum-optics is the fact that squeezing implies a *Bogoliubov transformation* (linear canonical transformation) that mixes creation and annihilation operators in the Heisenberg picture [39]:

$$\hat{S}^\dagger(\zeta) \hat{a} \hat{S}(\zeta) = \hat{a} \cosh r - e^{i\theta} \hat{a}^\dagger \sinh r. \quad (2.5)$$

Such mixing is the operator level source of the smaller quantum fluctuations: this mixing creates *anomalous* (pair) correlations of the form $\langle \hat{a}\hat{a} \rangle$ and $\langle \hat{a}^\dagger \hat{a}^\dagger \rangle$ which vanish for coherent and thermal states. For the squeezed vacuum, the bosonic moments satisfy the well-known relations.

$$\langle \hat{a}^\dagger \hat{a} \rangle = \sinh^2 r, \quad \langle \hat{a}\hat{a} \rangle = -\frac{1}{2} e^{i\theta} \sinh(2r), \quad (2.6)$$

it is important to point out that the squeezer is coded directly on the level of second-order operator moments [39, 11]. Because Gaussian states are completely characterized by first and second moments, this bosonic moment structure provides an efficient route for treating squeezed states in linear optical networks and open-system models [11, 12].

Physically-platform wise, squeezed states are usually created by using parametric (quadratic) interactions which are naturally bosonic. Typical examples would be degenerate parametric amplification and four-wave mixing, whose good Hamiltonians include terms which are proportional to $\hat{a}^{\dagger 2}$ and $\hat{a}^2$ and therefore directly implement the generator appearing in $\hat{S}(\zeta)$ [39]. In integrated photonics and waveguide settings, such bosonic operator descriptions are particularly convenient because propagation, evanescent coupling, and loss can all be modeled as linear transformations and damping channels acting on $\hat{a}_j$ and $\hat{a}_j^\dagger$, allowing one to obtain closed equations for moments such as $\langle \hat{a}_j^\dagger \hat{a}_k \rangle$ and $\langle \hat{a}_j \hat{a}_k \rangle$ [11].

Two-Mode Squeezing and Operator Correlations Beyond single-mode squeezing, the bosonic literature also highlights the importance of *two-mode squeezed states*, generated by the two-mode squeezing operator

$$\hat{S}_{12}(\zeta) = \exp\left(\zeta^* \hat{a}_1 \hat{a}_2 - \zeta \hat{a}_1^\dagger \hat{a}_2^\dagger\right), \quad (2.7)$$

which creates strong intermodal pair correlations such as $\langle \hat{a}_1 \hat{a}_2 \rangle$ and $\langle \hat{a}_1^\dagger \hat{a}_2^\dagger \rangle$ [11, 12]. These anomalous cross moments are the operator-level signatures of EPR-type correlations and constitute the microscopic origin of Gaussian entanglement and steering when

the state is analyzed in the covariance-matrix framework [12].

Connection to covariance matrices Whereas the bosonic theory characterizes squeezing through transformations of operators and anomalous, it can be termed moments, and is entirely identical to the quadrature covariance-matrix description utilized to measure steering and Gaussian entanglement [11, 12]. Realistically, it is common to solve the dynamics in the bosonic basis to get second moments, and then projects these moments onto the quadrature covariance matrix that is the input. to engage and control in further analysis [11].

2.2 Covariance Matrix

In Continuous-variable (CV) quantum information, the covariance matrix serves as a fundamental tool for characterizing the properties of quantum states, particularly Gaussian states, which are completely described by their first and second statistical moments [13]. While Gaussian states exist in infinite-dimensional Hilbert spaces, the covariance matrix provides a technically accessible description using a finite number of degrees of freedom [12, 11].

The covariance matrix σ is a real, symmetric, and positive-definite matrix of size $2n \times 2n$ for a system of n bosonic modes [12, 11]. It represents the two-point correlation functions between canonical mode operators [12].

The covariance matrix for n bosonic modes is represented by

$$\sigma_{ij} = \frac{1}{2} \langle \hat{z}_i \hat{z}_j + \hat{z}_j \hat{z}_i \rangle - \langle \hat{z}_i \rangle \langle \hat{z}_j \rangle, \quad (2.8)$$

where $\hat{\mathbf{z}}$ is the phase-space vector and is defined as

$$\hat{\mathbf{z}} = (\hat{Q}_1, \hat{R}_1, \dots, \hat{Q}_n, \hat{R}_n)^T. \quad (2.9)$$

The degrees of freedom of the bosonic modes are represented by the quadrature operators $\hat{Q}_k$ and $\hat{R}_k$.

Here, σ is a real, symmetric matrix that must satisfy the quantum uncertainty principle in order to represent a physical Gaussian state [13].

For a two-mode Gaussian system consisting of modes A and B , the covariance matrix

can be written in block form as [12, 11]

$$\sigma_{MN} = \begin{pmatrix} \mathbf{M} & \mathbf{L} \\ \mathbf{L}^T & \mathbf{N} \end{pmatrix}, \quad (2.10)$$

Here, the diagonal blocks i.e. $\mathbf{M}$ and $\mathbf{N}$ are 2×2 real symmetric matrices describing only local noise and squeezing of each mode, whereas the off-diagonal blocks $\mathbf{L}$ and $\mathbf{L}^T$ are 2×2 real matrices containing all cross-quadrature moments and therefore fully encode the correlations between the two modes [12]. The covariance matrix in its standard form can be written as

$$\sigma_{MN} = \begin{pmatrix} a & 0 & c_1 & 0 \\ 0 & a & 0 & c_2 \\ c_1 & 0 & b & 0 \\ 0 & c_2 & 0 & b \end{pmatrix}. \quad (2.11)$$

The variances a and b , and the covariances c_1 and c_2 can be determined using the four local symplectic invariants. For example, the determinant of the covariance matrix σ_{MN} is given by $(ab - c_1^2)(ab - c_2^2)$, while the determinants of the local covariance matrices are $\det \mathbf{M} = a^2$ and $\det \mathbf{N} = b^2$, and the determinant of the correlation matrix is $\det \mathbf{L} = c_1 c_2$. Consequently, every arbitrary two-mode covariance matrix admits a unique standard form. [4, 11, 12]

Physicality Condition A matrix is said to be physically valid covariance matrix if and only if it satisfies the Robertson–Schrödinger uncertainty principle. This condition can be expressed as the matrix inequality [12, 11]

$$\sigma + i\Omega \geq 0, \quad (2.12)$$

where Ω is the symplectic form, defined as the direct sum of n matrices of the form [12, 11]

$$\Omega = \bigoplus_{k=1}^n \omega, \quad \omega = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \quad (2.13)$$

2.3 Optical Waveguides as Quantum Systems

An optical waveguide is a structure that guides light along a desired path. It consists of thin, high-refractive-index core layer positioned between low-refractive-index cladding material and the light is trapped within the core through total internal reflection. Photons pass through these waveguides in particular patterns known as modes [14].

2.3.1 Evanescent Overlap/Coupled Mode Theory

When light passes through a waveguide system, the electric field is mostly trapped in the high-refractive-index core. Yet, a part of the field can penetrate into the surrounding lower-index cladding. We call this decaying field as the evanescent field. If we place another waveguide in this region, i.e. the evanescent field reaches typically less than 1-2 wavelengths away, light can tunnel between these two waveguides. We call this transfer of energy through coupling of evanescent fields as evanescent coupling. The coupling strength between the waveguides depends on the separation distance between the waveguides, refractive index and mode shapes [15].

2.3.2 Loss and Decoherence In Real Waveguide Systems

In real optical waveguide system, the performance is undermined by two processes: classical optical power loss, attenuation α and quantum information loss, decoherence Γ . Exponential decay of optical intensity as light propagates is called attenuation while decoherence is the process in which a quantum system loses its correlation properties, i.e. superposition or entanglement due to interaction with the environment, results in losing the necessary phase relationship and limiting visibility of interference [16, 56, 15]. In quantum photonics the two problems are interconnected, classical optical loss (α) acts as a primary source of quantum decoherence. When a photon is lost from the waveguide system, its quantum state becomes coupled to an unobserved vacuum port which results in transferring quantum information to the environment irreversibly and thus decoheres the entanglement and other nonclassical quantum resources. This loss mechanism is as an amplitude damping channel [56, 17]

2.3.3 Relevance Of Optical Waveguides For Quantum Circuits/Photonics

The optical waveguides are highly necessary in quantum photonics since they have facilitated the researchers to replace large laboratory systems with small chips that can soon be used in quantum technologies. All structures are built on the chip and this enables the researchers to implement quantum logic gates like CNOT, Toffoli and Fredkin gate on a chip. They find application in building quantum computers, secure communication and quantum sensors. Low loss photonic materials like silicon, silicon nitride and lithium niobate are highly essential in this fact since optical waveguides are very important in the character of quantum information processing of integrated photonic circuits. Their nonlinear and optical properties can be optimally used in control of generation, manipulation, and detection of quantum states with very low propagation loss. When photon sources, beam splitters, phase shifters, and detectors are implemented on a single chip on a waveguide basis the resultant circuit is compact, stable, and scalable [17, 18].

2.4 Quantum Steering

The phenomenon known as quantum steering was introduced in 1935 [8]. It is a form of quantum correlation that has received considerable attention in quantum information theory, especially after its formalisation in 2007 [38]. Schrodinger first described this phenomenon as "magic" as the effect does not facilitate transmission of information, and it is the ability of one party (Alice) to measure or steer the wave function of another distant party [19]. One common notion in the modern theory is the inability of the conditional quantum states of one party (Bob) to be described by a local hidden state (LHS) model. A joint bipartite quantum state ρ_{AB} is said to be steerable if Bob is forced to believe that Alice is influencing his system through some form of action at a distance. Conversely, when (LHS) exists for the shared state, then such a bipartite system is referred to as an unsteerable state [19, 20]. Steering lies between entanglement and Bell nonlocality, that is, only steerable quantum correlations can be verified in one-sided device independent scenario, entanglement cannot [19, 20]. In Continuous-variable (CV) systems, generally implemented using optical field quadratures X and P , the steering-detection problem is often formulated in terms of criteria based on the Heisenberg uncertainty relation for conditional measurements. A state shared between A and B is steerable if the product of the minimum inferred variances of Bob's position (X_B) and momentum (P_B) quadratures, in terms of the measurements of Alice, is lower

than the quantum limit: [21, 19]

$$V_{\text{inf}}(X_B) V_{\text{inf}}(P_B) < 1. \quad (2.14)$$

For CV Gaussian states where both the states and measurements are Gaussian, this EPR criterion is both necessary and sufficient to detect bipartite steering[21]. Remarkably, a bipartite Gaussian state is unsteerable with Gaussian measurements if and only if a specific matrix inequality involving the covariance matrix V_{AB} is satisfied:

$$V_{AB} + i(0_A \oplus \Omega_B) \geq 0. \quad (2.15)$$

If a quantum state is steerable with Gaussian measurements, it is also steerable with a certain pair of canonical quadratures [19].

2.4.1 Quantification Of Quantum Steering Using Gaussian Steering Measure

The Gaussian $1 \rightarrow 2$ steerability $\mathcal{S}_{1 \rightarrow 2}(\sigma_{MN})$ is defined based on the symplectic eigenvalues $\{\bar{\nu}_j^N\}$ of the Schur complement matrix M_{σ_N} , calculated from the state's covariance matrix σ_{MN} :

$$\mathcal{S}_{1 \rightarrow 2}(\sigma_{MN}) := \max \left\{ 0, - \sum_{\bar{\nu}_j^N < 1} \ln(\bar{\nu}_j^N) \right\}. \quad (2.16)$$

For the specific case of two-mode Gaussian states (where the steered party has only one mode), the measure simplifies to the Rényi-2 coherent information:

$$\mathcal{S}_{1 \rightarrow 2}(\sigma_{MN}) = \max \left\{ 0, \frac{1}{2} \ln \left(\frac{\det \sigma_M}{\det \sigma_{MN}} \right) \right\} = \max \{0, H_2(\sigma_M) - H_2(\sigma_{MN})\}, \quad (2.17)$$

where $H_2(\sigma)$ denotes the Rényi-2 entropy [22].

Properties and Hierarchy: The Gaussian steering measure satisfies several desirable mathematical properties: it is invariant under local unitaries, convex, additive, and monotonically non-increasing under quantum operations performed on the untrusted party (Alice). It vanishes if and only if the state is nonsteerable by Gaussian measure-

ments. Moreover, it respects the hierarchy of correlations:

$$\mathcal{S}_{1 \rightarrow 2}(\sigma_{MN}) \leq \mathcal{E}_2(\sigma_{MN}), \quad (2.18)$$

where $\mathcal{E}_2$ denotes the Gaussian Rényi-2 measure of entanglement. For pure states,

$$\mathcal{S}_{1 \rightarrow 2}(\sigma_{MN}^p) = \mathcal{E}_2(\sigma_{MN}^p). \quad (2.19)$$

Operational Meaning: For two-mode Gaussian states, Gaussian steerability has a direct operational interpretation in one-sided device-independent QKD:

$$\mathcal{K} \geq \max \{0, \mathcal{S}_{2 \rightarrow 1}(\sigma_{MN}) + \ln 2 - 1\}, \quad (2.20)$$

where $\mathcal{K}$ is the guaranteed secure key rate [22].

2.5 Entanglement In Continous Variable Systems

Entanglement is fundamental property of a quantum theory, and it is a central resource for most quantum information science and emerging quantum technologies [23]. Originating from conceptual structures identified by Einstein, Podolsky, and Rosen (EPR) and Schrödinger, entanglement describes nonlocal correlations within composite quantum systems that cannot be replicated by classical systems [25]. It is a physical fact that two or more particles show correlations which cannot be explained classically, and these particles will be correlated no matter how far they are from each other [1, 25]. Historically considered a "spooky" phenomenon, entanglement is now an important part of engineering essential quantum protocols such as quantum computing and quantum teleportation in CV systems associated with infinite dimensional Hilbert spaces, provide the advantage of allowing deterministic and unconditional quantum operations of which is often a shortcoming in DV schemes due to relying on an imperfect single-photon generation and detection [23]. Within the CV framework, the family of Gaussian states is the most applicable one, and it is the subject of entanglement research [26]. The quantitative way to determine the intensity or the extent of entanglement within a quantum system is termed as entanglement quantification. Quantification is needed and important because the measured degree of entanglement directly influences the performance of quantum information tasks [1, 25, 33].

2.5.1 Entanglement Quantification Measures

To measure the entanglement created and maintained in a system in a quantitative way, scientists use appropriate measures[1] . One of the most commonly used measure is logarithmic negativity E_N [57, 58].

E_N is defined as

$$E_N(t) = \log_2 \left\| \rho_{MN}^{T_N} \right\|_1, \quad (2.21)$$

where ρ_{MN} denotes the density matrix of the bipartite system M and N , $\rho_{MN}^{T_N}$ denotes the partial transpose with respect to subsystem N , and $\| \cdot \|_1$ represents the trace norm. A non-zero value for E_N confirms that the state is entangled.

For Gaussian states, and sub Gaussian states such as squeezed states, the measure of entanglement, E_N , is determined by the symplectic eigenvalues ($\tilde{\nu}$) of the partially transposed covariance matrix ($\tilde{\sigma}_{MN}$). The state is entangled if the smallest symplectic eigenvalue, $\tilde{\nu}$ satisfies

$$\tilde{\nu} < \frac{1}{2}. \quad (2.22)$$

Logarithmic negativity in this context is given by

$$E_N(t) = \max [0, -\ln (2 \tilde{\nu}(t))] . \quad (2.23)$$

and serves as an upper bound to the distillable entanglement [57, 58, 28] .

2.6 Loss and Decoherence

Decoherence has become a prominent subject in modern physics, It is motivated by the desire to understand the border between the classical and quantum world, and the manner in which it is detrimental to the processing of quantum information [29]. Fundamentally, decoherence is nothing but the irreversible and uncontrollable accumulation of quantum correlations, such as entanglement and steering, between a system and its environment [29, 30]. Decoherence has attracted a lot of interest because of its relevance in explaining the quantum-classical boundary as well as its harmful effects on quantum information processing technologies. The main mechanism is that the environment actively monitors some system observables and hence, the decoherence of non-preferred states takes place. This monitoring thus imposes the choice of pointer states,

which are robust against environmental interactions[31, 29]. These pointer states continue to correlate with their environment while remaining stable under environmental interactions. The rapid conversion of quantum superpositions to mixtures is the reason why superposition states, which are commonly known as "Schrodinger cat" states, do not seem to appear. The selection of such preferred states is known as Environment-Induced Superselection (Einselection). Einselection places a basic limitation on the bulk of the system's Hilbert space, causing it to be restricted to a small classical domain that is made up of pointer states [31].

2.6.1 Loss (Dissipation) and Pure Decoherence

Decoherence is usually accompanied by dissipation, which refers to the net exchange of energy with the environment [29]. But there are certain theoretical cases that focus on pure decoherence (or dephasing), in which there is no energy exchange with the environment. The phenomenon of pure decoherence is of special interest in explaining why macroscopic superpositions do not occur, since the mechanism that leads to this is proposed to rely on the distinguishability of quantum states rather than on the differences in their energy [29]. Pure decoherence can occur when the system Hamiltonian (H_S) approximately commutes with the system-bath interaction Hamiltonian (H_{int}), or when the system initial states vary at a rate significantly slower than the decoherence timescale. Pure decoherence is considered very important in theoretical models which attempt to explain the apparent lack of superpositions of macroscopically distinguishable states, including Schrodinger-cat states [29]. The roles of these phenomena should be differentiated: relaxation and noise (loss) are the processes in which the environment perturbs the system, decoherence and einselection are the processes in which the system perturbs the environment [31]. Another extremely significant thing about decoherence, particularly in macroscopic systems, is the separation of the time scales between decoherence and energy loss (relaxation). The rate at which a system decoheres is often depends on the separation between the superposed, macroscopically distinct states. In such cases, the dimensionless distance D can be very large (e.g., 10^{20}), so decoherence can be very fast, even when the rate of energy relaxation of the system is very small [32].

2.6.2 Loss Mechanism In Waveguides

Loss mechanism (κ) in waveguides is due to the loss in the material of the waveguide structure itself: Waveguides have often been designed, to be identical, and so the loss rate for the field modes (a_1 and a_2) is usually taken to be the same. Loss is theoretically modelled in the framework of system–reservoir interaction by using a quantum–Liouville equation of motion. This is the equation of evolution of the density operator (ρ) with two terms, the Hamiltonian term to the unitary coupling and the loss term ($L\rho$) to represent the dissipation:

$$\dot{\rho} = -\frac{i}{\hbar}[H, \rho] + L\rho, \quad (2.24)$$

where the loss operator $L\rho$ specifically represents the decay rates of the modes a_1 and a_2 [28, 56, 54, 55] .

2.7 Applications and Motivations

In this section, we discuss the applications and motivations of quantum entanglement, quantum steering, and optical waveguides.

2.7.1 Applications of Quantum Entanglement

The concept of quantum entanglement is a central pillar of quantum information theory and quantum mechanics and is used to explain highly correlated particles that become linked regardless of the distance separating them [1, 25]. This has a critical role in proving the theoretical and algorithmic superiority of quantum computers over its classical counterparts [34]. An important element of quantum computing is the manipulation of entanglement that results in the enhancement of the capabilities [34], and is applied in fundamental quantum information processing systems like quantum cryptography, quantum teleportation and quantum computing [1]. The quantification of entanglement is important in the determination of the intactness and the depth of the level of correlation that has a direct impact on the execution of quantum information tasks. Quantified entanglement measures include concurrence, negativity (Including logarithmic negativity), Schmidt decomposition, and entanglement of formation [25]. In case of bipartite Gaussian states, entanglement can be quantitatively measured by the logarithmic negativity of the input states, such as squeezed states and photon number states, and the entanglement dynamics is quantitatively measured in terms of logarithmic negativity.

In waveguide architecture, the sustainability of generated entanglement can be significantly important in practical implementation of quantum communication and computation tasks, and dynamics of entanglement is studied by using quantitative measure known as logarithmic negativity [28]. Highly entangled states are essential in doing complex quantum information processing and implementing error-correcting code [34]. In short quantum entanglement is essential for achieving quantum computational advantage and quantum error-correcting codes, it allows safe quantum communication protocols like quantum cryptography and quantum teleportation, it allows quantum information processing with Gaussian states, and it allows scalable on-chip implementations in integrated optical waveguide architectures [1, 25, 34, 11, 23, 28].

2.7.2 Applications Of Quantum Steering

Quantum steering is a form of quantum correlation that has been introduced by Schrodinger which allows one party, Alice, to change the state of a remote party, Bob, remotely, by acting on local measurements which make use of common entanglement[8, 38, 19]. One of the most practical uses of quantum steering is the realization of desired quantum states, the so-called Measurement-Induced Quantum Steering (MIQS), a protocol that can be used to prepare a target quantum state regardless of the initial (mixed) state, in other words, the preparation of quantum states [34]. Quantum Steering plays an important role in offering security to one-sided device-independent Quantum Key Distribution (1SDI QKD) protocols, where the amount of steerability (Gaussian) can be used to measure the rate of guaranteed key rate [22]. Quantum steering also has a close connection with quantum teleportation, any gaussian state that can perform safe teleportation must be two way steerable [35]. Quantum steering also plays an important role in quantum-enhanced metrology [36]. Hence, quantum steering has important applications in measurement-induced quantum state preparation, one-sided device-independent quantum key distribution, secure continuous-variable quantum teleportation, quantum cryptography, and quantum-enhanced metrology, where steerable correlations can be used to accomplish tasks that cannot be certified using entanglement alone [38, 19, 22, 35, 36].

2.7.3 Applications Of Waveguides

Coupled waveguide designs are significant fundamental components that scientists examine whether they can be employed as straightforward building blocks for quantum

circuits in integrated photonic devices. These are discrete optical systems that are quite efficient in controlling the movement of light. Waveguides are useful in executing key activities in quantum computing. They allowed a CNOT gate on a single silicon chip. such architectures form a basis for universal photonic quantum circuits and are relevant for the development of photonic implementations of quantum algorithms, including circuits related to Shor's quantum factoring algorithm. Beyond computing, waveguide arrays are used to investigate coherent phenomena such as quantum walks and a discrete form of the Talbot effect [28, 59, 46, 60]. In short, with integrated optical waveguides in place it is now possible to implement quantum logic gates and universal photonic quantum circuits, they give rise to on-chip implementations of quantum algorithms, including Shor-factoring, and enable quantum simulation by coherent phenomena such as quantum walks and Talbot effects, they also provide a scalable environment to study the propagation and stability of nonclassical states in realistic lossy and open quantum systems [59, 46, 60, 28].

Chapter 3

Model System and Theoretical Formulations

3.1 Physical Model and Assumptions

3.1.1 Physical System Configuration

Our system consist of two identical single mode waveguides which are placed parallel to each other in a close distance such that their evanescent feild overlap and enable coupling between them. The optical modes in each waveguide are represented by creation and anihilation operator a_1 and a_2 where a_1 represents optical mode in waveguide a_1 and a_2 represents optical modein waveguide a_2 .Each waveguide mode act as a bosonic harmonic oscillator. The two waveguides are assumed to be identical and at the same resonance frequency, and loss rate, which is a realistic assumption.

3.2 Hamiltonian Description of the Coupled Waveguide System

In this section, we discuss our system hamiltonian, and coupling between waveguide modes

3.2.1 System Hamiltonian

The hamiltonian for our system is given by

$$H = \hbar\omega \left(a_1^\dagger a_1 + a_2^\dagger a_2 \right) + \hbar J \left(a_1^\dagger a_2 + a_2^\dagger a_1 \right). \quad (3.1)$$

Here, a_1 and a_2 are the bosonic field operators for the optical modes in each waveguide, satisfying the commutation relations $[a_1, a_1^\dagger] = 1$ and $[a_2, a_2^\dagger] = 1$.

the symbol ω is angular frequency of the each waveguide mode and the parameter J in the second part of the hamiltonian represents the coupling strength between the two waveguides and its value depends on the seperation distance between the waveguides.

The hamiltonian consists of two main parts:

Free energy terms: correspond to the free energy of the individual waveguide modes,

$$\hbar\omega \left(a_1^\dagger a_1 + a_2^\dagger a_2 \right).$$

Coupling terms: account for the evanescent coupling between the two waveguide modes,

$$\hbar J \left(a_1^\dagger a_2 + a_2^\dagger a_1 \right).$$

This term is responsible for the transfer of energy between the two waveguide modes and is the main focus of interest in this work.

3.2.2 Physical Interpretation of the Coupling

The evanescent interaction between the two parallel optical waveguides is represented by the coupling term in the system Hamiltonian. When the waveguides are sufficiently close to each other, the electromagnetic field of the guided mode extends beyond the core of the waveguide and overlaps with the field of the neighboring waveguide. This overlap allows the coherent exchange of photons between the two modes and is called as evascent overlapping [15, ?].

The strength of the coupling between the waveguides is defined by the parameter J which defines the speed with which energy moves between the waveguides. The higher the value of J the higher the mode overlap and higher the speed of oscillatory exchange of photons. when $J=0$ two entirely waveguides are independent and are considered uncoupled. Physically, such an interaction is similar to a beam splitter, where the photons coherently and simultaneously hopped between the two modes without loss in ideal case [15, 28, 46].

Without the dissipation, the coherent exchange between the two waveguides results in a periodic oscillation of photon betwenen the two waveguides. In case of losses, such oscillations are slowly damped, although the coupling term still plays the role of creating and maintaining quantum correlations. The coupling is therefore the core of creation maintenance of quantum interactions in integrated photonic systems [28, 15].

3.3 Open-System Dynamics and Loss Modelling

In realistic integrated photonic systems, optical modes experience losses. The coupled waveguide system cannot be treated as an isolated system, it needs to be formulated in the context of an open quantum systems. To include these dissipative effects, the system dynamics is given by using a density-matrix functional, in which the interaction with the external environment results in decoherence and energy decay [56, 28] . The time

evolution of the coupled waveguide system is in this section in terms of the Lindblad form of the quantum Liouville (master) equation. This approach allows inclusion of photon loss without changing the nature of the Gaussian states. This preservation is necessary for the analysis of quantum steering and entanglement [56, 54, 11, 28].

3.3.1 Quantum Liouville (Master) Equation

we can determine how this system evolve with loss by quantum liouville equation.

$$\dot{\rho} = -\frac{i}{\hbar}[H, \rho] + \mathcal{L}\rho \quad (3.2)$$

this equation is know as quantum liouville equation, it has two parts [56, 54]:

The Commutator:

$$[H, \rho]$$

which generates time evolution and is responsible for oscillatory exchange of energy and quantum correlation between the waveguides due to J.

Lindblad($L\rho$) term: deals with material loss, decoherence and decays due to interaction with the environment [56, 28].

3.3.2 Lindblad Operators for Waveguide Loss

$$\begin{aligned} \mathcal{L}\rho = & -k_1 \left(a_1^\dagger a_1 \rho + \rho a_1^\dagger a_1 - 2a_1 \rho a_1^\dagger \right) \\ & - k_2 \left(a_2^\dagger a_2 \rho + \rho a_2^\dagger a_2 - 2a_2 \rho a_2^\dagger \right), \end{aligned} \quad (3.3)$$

k_1 and k_2 represents the loss rates of modes a_1 and a_2 , and $a_1, a_1^\dagger$ are the creation and annihilation operators for mode a_1 , while ρ is the density matrix. a_2 and $a_2^\dagger$ are the creation and annihilation operators for mode a_2 [56, 28].

3.4 Dynamical Equations of the Model System

This section deals with the derivation of the dynamical equations of the coupled waveguide system. Through the calculation of expectation values, the open-system dynamics may be defined as a closed set of second-order moments, which is sufficient to describe the Gaussian states.

3.4.1 Expectation-Value Formalism

The time evolution of any system observable O is obtained from [56]

$$\frac{d}{dt}\langle O \rangle = \text{Tr}(O\dot{\rho}) = -i\langle [O, H] \rangle + \langle \mathcal{L}^\dagger(O) \rangle. \quad (3.4)$$

3.4.2 Definition of Relevant Second-Order Moments

$$n_1(t) = \langle a_1^\dagger(t) a_1(t) \rangle, \quad (3.5)$$

$$n_2(t) = \langle a_2^\dagger(t) a_2(t) \rangle, \quad (3.6)$$

$$m_1(t) = \langle a_1(t) a_1(t) \rangle, \quad (3.7)$$

$$m_2(t) = \langle a_2(t) a_2(t) \rangle, \quad (3.8)$$

$$c(t) = \langle a_1^\dagger(t) a_2(t) \rangle, \quad (3.9)$$

$$s(t) = \langle a_1(t) a_2(t) \rangle. \quad (3.10)$$

Here, $n_1(t)$ and $n_2(t)$ represent the mean photon numbers in the first and second waveguides, respectively. The moments $m_1(t)$ and $m_2(t)$ describe single-mode squeezing correlations, while $c(t)$ and $s(t)$ quantify inter-mode correlations responsible for quantum entanglement and steering between the two waveguides. to be written)

3.5 Moment Equations of Motion

in this section we discuss the moment equations we have got after

3.5.1 Coupled Differential Equations

Using $\frac{d}{dt}\langle O \rangle = \text{Tr}(O\dot{\rho})$, the equations of motion for the relevant second-order moments are obtained as

$$\frac{d}{dt}\langle a_1^\dagger a_1 \rangle = iJ\left(\langle a_2^\dagger a_1 \rangle - \langle a_1^\dagger a_2 \rangle\right) - k_1\langle a_1^\dagger a_1 \rangle, \quad (3.11)$$

$$\frac{d}{dt}\langle a_2^\dagger a_2 \rangle = iJ\left(\langle a_1^\dagger a_2 \rangle - \langle a_2^\dagger a_1 \rangle\right) - k_2\langle a_2^\dagger a_2 \rangle, \quad (3.12)$$

$$\frac{d}{dt}\langle a_1^\dagger a_2 \rangle = iJ\left(\langle a_2^\dagger a_2 \rangle - \langle a_1^\dagger a_1 \rangle\right) - \frac{k_1 + k_2}{2}\langle a_1^\dagger a_2 \rangle, \quad (3.13)$$

$$\frac{d}{dt}\langle a_2^\dagger a_1 \rangle = iJ\left(\langle a_1^\dagger a_1 \rangle - \langle a_2^\dagger a_2 \rangle\right) - \frac{k_1 + k_2}{2}\langle a_2^\dagger a_1 \rangle, \quad (3.14)$$

$$\frac{d}{dt}\langle a_1 a_1 \rangle = -2iJ\langle a_1 a_2 \rangle - k_1\langle a_1 a_1 \rangle, \quad (3.15)$$

$$\frac{d}{dt}\langle a_2 a_2 \rangle = -2iJ\langle a_1 a_2 \rangle - k_2\langle a_2 a_2 \rangle, \quad (3.16)$$

$$\frac{d}{dt}\langle a_1 a_2 \rangle = -iJ\left(\langle a_1 a_1 \rangle + \langle a_2 a_2 \rangle\right) - \frac{k_1 + k_2}{2}\langle a_1 a_2 \rangle. \quad (3.17)$$

These equations form a closed set due to the quadratic form of the Hamiltonian and the Gaussian-preserving nature of the Lindblad dissipation.

3.5.2 Reduction of Independent Variables

Although seven coupled moment equations are obtained, not all of them are independent. The inter-mode coherence moments satisfy the Hermitian relation

$$\langle a_2^\dagger a_1 \rangle = \langle a_1^\dagger a_2 \rangle^*. \quad (3.18)$$

As a result, the system dynamics can be fully described by six independent second-order moments, while the remaining coherence follows directly from this relation.

3.6 Analytical Solutions of the Moment Equations

3.6.1 Equal-Loss Approximation

For identical waveguides, we assume equal loss rates,

$$k_1 = k_2 = \kappa. \quad (3.19)$$

3.6.2 General Analytical Solutions

Without imposing any restriction on the initial correlations, the exact solutions of the moment equations are given by

$$\begin{aligned} \langle a_1^\dagger a_1 \rangle(t) = & \frac{e^{-\kappa t}}{2} \left[\langle a_1^\dagger a_1 \rangle_0 + \langle a_2^\dagger a_2 \rangle_0 + (\langle a_1^\dagger a_1 \rangle_0 - \langle a_2^\dagger a_2 \rangle_0) \cos(2Jt) \right. \\ & \left. + i(\langle a_2^\dagger a_1 \rangle_0 - \langle a_1^\dagger a_2 \rangle_0) \sin(2Jt) \right], \end{aligned} \quad (3.20)$$

$$\begin{aligned} \langle a_2^\dagger a_2 \rangle(t) = & \frac{e^{-\kappa t}}{2} \left[\langle a_1^\dagger a_1 \rangle_0 + \langle a_2^\dagger a_2 \rangle_0 - (\langle a_1^\dagger a_1 \rangle_0 - \langle a_2^\dagger a_2 \rangle_0) \cos(2Jt) \right. \\ & \left. - i(\langle a_2^\dagger a_1 \rangle_0 - \langle a_1^\dagger a_2 \rangle_0) \sin(2Jt) \right], \end{aligned} \quad (3.21)$$

$$\begin{aligned} \langle a_1^\dagger a_2 \rangle(t) = & e^{-\kappa t} \left[\frac{\langle a_1^\dagger a_2 \rangle_0 + \langle a_2^\dagger a_1 \rangle_0}{2} + \frac{\langle a_1^\dagger a_2 \rangle_0 - \langle a_2^\dagger a_1 \rangle_0}{2} \cos(2Jt) \right. \\ & \left. - \frac{i}{2} (\langle a_1^\dagger a_1 \rangle_0 - \langle a_2^\dagger a_2 \rangle_0) \sin(2Jt) \right], \end{aligned} \quad (3.22)$$

$$\begin{aligned} \langle a_2^\dagger a_1 \rangle(t) = & e^{-\kappa t} \left[\frac{\langle a_1^\dagger a_2 \rangle_0 + \langle a_2^\dagger a_1 \rangle_0}{2} - \frac{\langle a_1^\dagger a_2 \rangle_0 - \langle a_2^\dagger a_1 \rangle_0}{2} \cos(2Jt) \right. \\ & \left. + \frac{i}{2} (\langle a_1^\dagger a_1 \rangle_0 - \langle a_2^\dagger a_2 \rangle_0) \sin(2Jt) \right], \end{aligned} \quad (3.23)$$

$$\begin{aligned} \langle a_1 a_1 \rangle(t) = & \frac{e^{-\kappa t}}{2} \left[\langle a_1 a_1 \rangle_0 - \langle a_2 a_2 \rangle_0 + (\langle a_1 a_1 \rangle_0 + \langle a_2 a_2 \rangle_0) \cos(2Jt) - 2i \langle a_1 a_2 \rangle_0 \sin(2Jt) \right], \end{aligned} \quad (3.24)$$

$$\begin{aligned} \langle a_2 a_2 \rangle(t) = & \frac{e^{-\kappa t}}{2} \left[-(\langle a_1 a_1 \rangle_0 - \langle a_2 a_2 \rangle_0) + (\langle a_1 a_1 \rangle_0 + \langle a_2 a_2 \rangle_0) \cos(2Jt) - 2i \langle a_1 a_2 \rangle_0 \sin(2Jt) \right], \end{aligned} \quad (3.25)$$

$$\langle a_1 a_2 \rangle(t) = e^{-\kappa t} \left[\langle a_1 a_2 \rangle_0 \cos(2Jt) - \frac{i}{2} (\langle a_1 a_1 \rangle_0 + \langle a_2 a_2 \rangle_0) \sin(2Jt) \right]. \quad (3.26)$$

3.6.3 Special Case: Independently Prepared Input Modes

For independently prepared input modes, the initial inter-mode correlations vanish,

$$\langle a_1^\dagger a_2 \rangle_0 = \langle a_1 a_2 \rangle_0 = 0,$$

leading to simplified expressions that are used throughout the remainder of this thesis.

3.6.4 Compact Representation Using Gaussian Moments

For convenience, we give the following names to the moments:

$$n_1(t) = \langle a_1^\dagger a_1 \rangle(t), \quad n_2(t) = \langle a_2^\dagger a_2 \rangle(t),$$

$$c(t) = \langle a_1^\dagger a_2 \rangle(t), \quad m_1(t) = \langle a_1 a_1 \rangle(t), \quad m_2(t) = \langle a_2 a_2 \rangle(t), \quad s(t) = \langle a_1 a_2 \rangle(t).$$

Defining the constants

$$C_1 = \langle a_1^\dagger a_1 \rangle_0 + \frac{1}{2}, \quad C_2 = \langle a_2^\dagger a_2 \rangle_0 + \frac{1}{2},$$

$$D_1 = \langle a_1 a_1 \rangle_0, \quad D_2 = \langle a_2 a_2 \rangle_0,$$

The term $\frac{1}{2}$ accounts for the unavoidable vacuum (zero-point) fluctuations of the bosonic field.

The Gaussian moments evolve as:

$$\begin{aligned} n_1(t) &= \frac{e^{-\kappa t}}{2} [(C_1 + C_2 - 1) + (C_1 - C_2) \cos(2Jt)], \\ n_2(t) &= \frac{e^{-\kappa t}}{2} [(C_1 + C_2 - 1) - (C_1 - C_2) \cos(2Jt)], \\ c(t) &= \frac{i e^{-\kappa t}}{2} (C_2 - C_1) \sin(2Jt), \\ m_1(t) &= \frac{e^{-\kappa t}}{2} [(D_1 + D_2) \cos(2Jt) + (D_1 - D_2)], \\ m_2(t) &= \frac{e^{-\kappa t}}{2} [(D_1 + D_2) \cos(2Jt) + (D_2 - D_1)], \\ s(t) &= -\frac{i e^{-\kappa t}}{2} (D_1 + D_2) \sin(2Jt). \end{aligned}$$

These expressions are used directly in the numerical evaluations of Gaussian steering and entanglement throughout this thesis.

A single-mode Gaussian state is fully characterized by the parameters of its covariance matrix. The diagonal element c_j quantifies the local quadrature variance of the j -th mode and is directly related to the mean photon number according to

$$\langle a_j^\dagger a_j \rangle_0 = c_j - \frac{1}{2}, \tag{3.27}$$

where the term $\frac{1}{2}$ accounts for the unavoidable vacuum (zero-point) fluctuations of the bosonic field.

Within the nonclassicality–purity parametrization, the same quantity c_j can be expressed in terms of the nonclassicality τ_j and the purity μ_j as

$$c_j = \frac{\sqrt{\tau_j^2 - \tau_j + \frac{1}{(2\mu_j)^2} - \frac{1}{4}}}{1 - 2\tau_j} + \frac{1}{2}, \quad (3.28)$$

which guarantees that the corresponding covariance matrix satisfies the Heisenberg uncertainty principle.

The off-diagonal covariance parameter d_j is defined through the anomalous second-order moment

$$\langle a_j a_j \rangle_0 = -d_j, \quad (3.29)$$

where $d_j = |d_j|e^{i\phi_j}$ is, in general, a complex quantity. Within the same nonclassicality–purity parametrization, its magnitude is given by

$$|d_j| = \frac{\sqrt{\tau_j - \tau_j^2 + \frac{1}{(2\mu_j)^2} - \frac{1}{4}}}{1 - 2\tau_j}. \quad (3.30)$$

These two parameters, c_j and d_j , together fully specify the single-mode Gaussian input state, with c_j determining the phase-insensitive fluctuations and d_j encoding the phase-sensitive correlations of the field [37].

3.7 Covariance Matrix and Quantification of Quantum Steering and Entanglement

Moments obtained in this chapter are entirely sufficient to determine the quantum state of the system at all times.

The moments, $n_1(t)$, $n_2(t)$, $m_1(t)$, $m_2(t)$, $c(t)$, $s(t)$ calculates the covariance of a pair of modes optical field. This covariance matrix is the basic object of measuring the Gaussian quantum correlations. These moments are directly used to calculate using the covariance matrix formalism, the quantification of quantum steering and quantum entanglement. The analytic expressions derived in this chapter creates a connection be-

tween the physical parameters(strength of coupling, loss rate and initial state properties) of the waveguide system and the dynamical behavior of quantum correlations [10, 11]

3.7.0.1 Covariance Matrix of the Two-Mode System

The covariance matrix is given in complex operator form, $(a_1, a_1^\dagger, a_2, a_2^\dagger)$ as

$$\Upsilon(t) = \begin{pmatrix} \mathbf{M}(t) & \mathbf{L}(t) \\ \mathbf{L}^\dagger(t) & \mathbf{N}(t) \end{pmatrix}, \quad (3.31)$$

In this matrix, $\mathbf{M}$, $\mathbf{N}$ and $\mathbf{L}$ are 2×2 matrices, where $\mathbf{M}$ and $\mathbf{N}$ that represent the local variations in the first and second modes in the waveguides, and $\mathbf{L}$ is the matrix that represents the correlation between the modes. Explicitly, these blocks are given by

$$\mathbf{M} = \begin{pmatrix} \langle a_1^\dagger a_1 \rangle + \frac{1}{2} & -\langle a_1 a_1 \rangle \\ -\langle a_1 a_1 \rangle^* & \langle a_1^\dagger a_1 \rangle + \frac{1}{2} \end{pmatrix}, \quad \mathbf{N} = \begin{pmatrix} \langle a_2^\dagger a_2 \rangle + \frac{1}{2} & -\langle a_2 a_2 \rangle \\ -\langle a_2 a_2 \rangle^* & \langle a_2^\dagger a_2 \rangle + \frac{1}{2} \end{pmatrix}, \quad (3.32)$$

and

$$\mathbf{L} = \begin{pmatrix} \langle a_1 a_2^\dagger \rangle & -\langle a_1 a_2 \rangle \\ -\langle a_1 a_2 \rangle^* & \langle a_1 a_2^\dagger \rangle^* \end{pmatrix}. \quad (3.33)$$

The covariance matrix in Eq. (3.31) fully characterizes the Gaussian state itself, and it offers the means of identifying different quantum correlations. [10, ?, 12].

This block-structured covariance matrix is the starting point for evaluating both Gaussian entanglement and quantum steering.

3.8 Gaussian Quantum Steering

Gaussian steering is determined by building the conditional covariance matrix of one mode conditional on measurements made on the other mode. Mathematically, we do it by the Schur complement of the total covariance. The conditional covariance matrix to transform mode 1 mode 2 is.

$$\Upsilon_{2|1} = \mathbf{N} - \mathbf{L}^\dagger \mathbf{M}^{-1} \mathbf{L}. \quad (3.34)$$

The degree of Gaussian steering from mode 1 to mode 2 is quantified by [22].

$$S_{1 \rightarrow 2} = \max \left[0, \frac{1}{2} \log_2 \frac{\det \mathbf{N}}{\det \mathbf{\Upsilon}_{2|1}} \right]. \quad (3.35)$$

3.8.1 Gaussian Entanglement

With the covariance matrix we can detect the Gaussian entanglement by performing the partial transposition operation, which is the laxity of the time of one part of the system in phase space. This can easily be expressed in the form of the smallest symplectic. eigenvalue $\tilde{\nu}_-$ of the partially transposed covariance matrix of eigenvalue and the logarithmic negativity(E_N) is the measure of the degree of entanglement.

$$E_N = \max[0, -\log_2(2\tilde{\nu}_-)]. \quad (3.36)$$

This applies to both subsystems and provides a test that is necessary and sufficient to test bipartite Gaussian entanglement [28].

For a two-mode Gaussian state, the smallest symplectic eigenvalue $\tilde{\nu}_-$ of the partially transposed covariance matrix is given by

$$\tilde{\nu}_- = \sqrt{\frac{\tilde{\Delta} - \sqrt{\tilde{\Delta}^2 - 4 \det \boldsymbol{\sigma}}}{2}},$$

where

$$\tilde{\Delta} = \det \mathbf{A} + \det \mathbf{B} - 2 \det \mathbf{C}.$$

[28]

Covariance-matrix representation and PPT criterion. In continuous-variable Gaussian systems, bipartite entanglement can be fully characterized at the level of second-order moments through the covariance matrix $\boldsymbol{\sigma}$. In this work, the covariance matrix of the two-mode Gaussian state is naturally constructed from the field operators in the complex operator basis $(a_1, a_1^\dagger, a_2, a_2^\dagger)$, as it directly follows from the Heisenberg equations of motion for the coupled waveguide modes. The corresponding real quadrature operators are defined as

$$\hat{Q}_j = \frac{1}{\sqrt{2}}(a_j + a_j^\dagger), \quad \hat{R}_j = \frac{i}{\sqrt{2}}(a_j^\dagger - a_j),$$

which satisfy the canonical commutation relation $[\hat{Q}_j, \hat{R}_k] = i\delta_{jk}$ and are related to the operator basis through a linear symplectic transformation.

Although the covariance matrix is expressed in the operator basis, all quantities relevant for entanglement detection depend only on symplectic invariants. In particular, the determinants of the block matrices A , B , and C , as well as $\det \sigma$, remain invariant under symplectic transformations between the operator and quadrature representations [13, 10]. Consequently, the logarithmic negativity can be evaluated directly using the positive partial transpose (PPT) criterion, which for Gaussian states reduces to the condition that the smallest symplectic eigenvalue $\tilde{\nu}_<$ of the partially transposed covariance matrix satisfies $\tilde{\nu}_< < \frac{1}{2}$. The corresponding PPT symplectic invariant is

$$\tilde{\Delta} = \det A + \det B - 2 \det C,$$

which is algebraically equivalent to performing partial transposition in the quadrature representation. This approach is standard in the analysis of Gaussian entanglement and is widely employed in continuous-variable quantum optics [40, 26, 28].

3.8.2 Relation Between Entanglement and Steering

Despite the fact that logarithmic negativity and Gaussian steering are derived from the same covariance matrix, they explore different aspects of quantum correlations. Entanglement is detected by the partial transposition of the covariance matrix and is symmetric between the subsystems, whereas steering is detected through conditional uncertainty relation and is intrinsically directional. All the steerable Gaussian states are therefore entangled, but not all entangled states exhibit steering [19, 38].

The hierarchy relation between quantum correlations is For continuous-variable Gaussian states, the same strict hierarchy holds [38]:

$$\text{Bell nonlocality} \subsetneq \text{Gaussian Steering} \subsetneq \text{Gaussian Entanglement}. \quad (3.37)$$

For our Gaussian states, the hierarchy reduces to

$$\text{Gaussian Steering} \subsetneq \text{Gaussian Entanglement}. \quad (3.38)$$

it means quantum steering is a subset of quantum entanglement [19].

3.9 squeezed states

3.9.1 Squeezed covariance matrix

We define the bosonic operator vector as,

$$\hat{\mathbf{A}} = (\hat{a}_1, \hat{a}_1^\dagger, \hat{a}_2, \hat{a}_2^\dagger)^T, \quad [\hat{a}_j, \hat{a}_k^\dagger] = \delta_{jk}. \quad (3.39)$$

For a zero-mean Gaussian state, we collect the *raw* second moments in the bosonic basis as

$$M_A(\theta) = \langle \hat{\mathbf{A}} \hat{\mathbf{A}}^T \rangle, \quad (3.40)$$

so that each entry is of the form $[M_A(\theta)]_{ij} = \langle \hat{A}_i \hat{A}_j \rangle$.

In the ordered basis $(\hat{a}_1, \hat{a}_1^\dagger, \hat{a}_2, \hat{a}_2^\dagger)$, the second-moment matrix can be written explicitly as

$$M_A(\theta) = \begin{pmatrix} \langle \hat{a}_1 \hat{a}_1 \rangle & \langle \hat{a}_1 \hat{a}_1^\dagger \rangle & \langle \hat{a}_1 \hat{a}_2 \rangle & \langle \hat{a}_1 \hat{a}_2^\dagger \rangle \\ \langle \hat{a}_1^\dagger \hat{a}_1 \rangle & \langle \hat{a}_1^\dagger \hat{a}_1^\dagger \rangle & \langle \hat{a}_1^\dagger \hat{a}_2 \rangle & \langle \hat{a}_1^\dagger \hat{a}_2^\dagger \rangle \\ \langle \hat{a}_2 \hat{a}_1 \rangle & \langle \hat{a}_2 \hat{a}_1^\dagger \rangle & \langle \hat{a}_2 \hat{a}_2 \rangle & \langle \hat{a}_2 \hat{a}_2^\dagger \rangle \\ \langle \hat{a}_2^\dagger \hat{a}_1 \rangle & \langle \hat{a}_2^\dagger \hat{a}_1^\dagger \rangle & \langle \hat{a}_2^\dagger \hat{a}_2 \rangle & \langle \hat{a}_2^\dagger \hat{a}_2^\dagger \rangle \end{pmatrix}. \quad (3.41)$$

For the lossy separable two-mode squeezed input considered here, the only nonzero anomalous moments are $\langle \hat{a}_j \hat{a}_j \rangle$ and the coupling-induced intermodal moment $\langle \hat{a}_1 \hat{a}_2 \rangle$.

In particular, the following mixed creation–annihilation cross correlations vanish:

$$\langle \hat{a}_1 \hat{a}_2^\dagger \rangle = \langle \hat{a}_2 \hat{a}_1^\dagger \rangle = \langle \hat{a}_1^\dagger \hat{a}_2 \rangle = \langle \hat{a}_2^\dagger \hat{a}_1 \rangle = 0, \quad (3.42)$$

so that Eq. (3.41) reduces to

$$M_A(\theta) = \begin{pmatrix} \langle \hat{a}_1 \hat{a}_1 \rangle & \langle \hat{a}_1 \hat{a}_1^\dagger \rangle & \langle \hat{a}_1 \hat{a}_2 \rangle & 0 \\ \langle \hat{a}_1^\dagger \hat{a}_1 \rangle & \langle \hat{a}_1^\dagger \hat{a}_1^\dagger \rangle & 0 & \langle \hat{a}_1^\dagger \hat{a}_2^\dagger \rangle \\ \langle \hat{a}_2 \hat{a}_1 \rangle & 0 & \langle \hat{a}_2 \hat{a}_2 \rangle & \langle \hat{a}_2 \hat{a}_2^\dagger \rangle \\ 0 & \langle \hat{a}_2^\dagger \hat{a}_1^\dagger \rangle & \langle \hat{a}_2^\dagger \hat{a}_2 \rangle & \langle \hat{a}_2^\dagger \hat{a}_2^\dagger \rangle \end{pmatrix}. \quad (3.43)$$

The remaining entries follow from Hermitian conjugation together with bosonic com-

mutation relations, in particular

$$\langle \hat{a}_j \hat{a}_j^\dagger \rangle = \langle \hat{a}_j^\dagger \hat{a}_j \rangle + 1. \quad (3.44)$$

Chapter 4

Results and discussion

In this chapter, we studied the dynamical behaviour of the Gaussian Quantum Steering and Gaussian Quantum Entanglement in a coupled lossy waveguide system using the Covariance matrix formalism. This analysis was conducted by initially considering a general reference case where the nonclassicality depth, purity and loss parameters were held constant and then by considering the action of each of these parameters separately. we investigated the effect of these parameters by keeping the coupling term J constant i.e 1 and by plotting Quantum Entanglement and Quantum Steering against $\frac{Jt}{\pi}$, where $J=1$. We also studied a special kind of gaussian states called squeezed states and analyzed the effect of squeezing on Quantum Entanglement and Quantum Steering. with the same consideration i.e $J=1$ and plotting Quantum Entanglement and Quantum Steering against $\frac{Jt}{\pi}$. we further investigated effect of coupling term J on Quantum Entanglement and Quantum Steering by keeping κ constant i.e 1, and plotted Quantum Entanglement and Quantum Steering against κt to see the effect.

4.1 Time Evolution of Gaussian Steering and Entanglement

This section discusses the analysis of the general time dynamics of the Gaussian Quantum Steering and Gaussian Entanglement in a coupled lossy waveguide system. The nonclassical depth τ , purity μ , and loss parameter κ are kept fixed, and time evolution dynamics of Entanglement and Steering is investigated. The evolution of the logarithmic negativity E_N and the Gaussian steering $\mathcal{S}$ with the normalized propagation time is illustrated by Fig 4.1. The quantum correlations are also periodic, which means the coherent exchange between quantum correlations of the two interacting waveguide modes. The maxima and minima of Gaussian steering and entanglement are situated in the same propagation lengths meaning that both the correlations can be related to the same underlying mode-coupling dynamics that is brought about by the evanescent interaction between the waveguides. It is also noted that the value of Gaussian steering is never greater than that of the Gaussian entanglement with time. This is agreeing with this fact of quantum correlations: steering is a more limited, but more effective form of nonclassicality than entanglement. As time passes, the steering and entanglement sluggishly reduce owing to the energy loss of the waveguides which signifies that the dissipation harms the conservation of the quantum correlations.

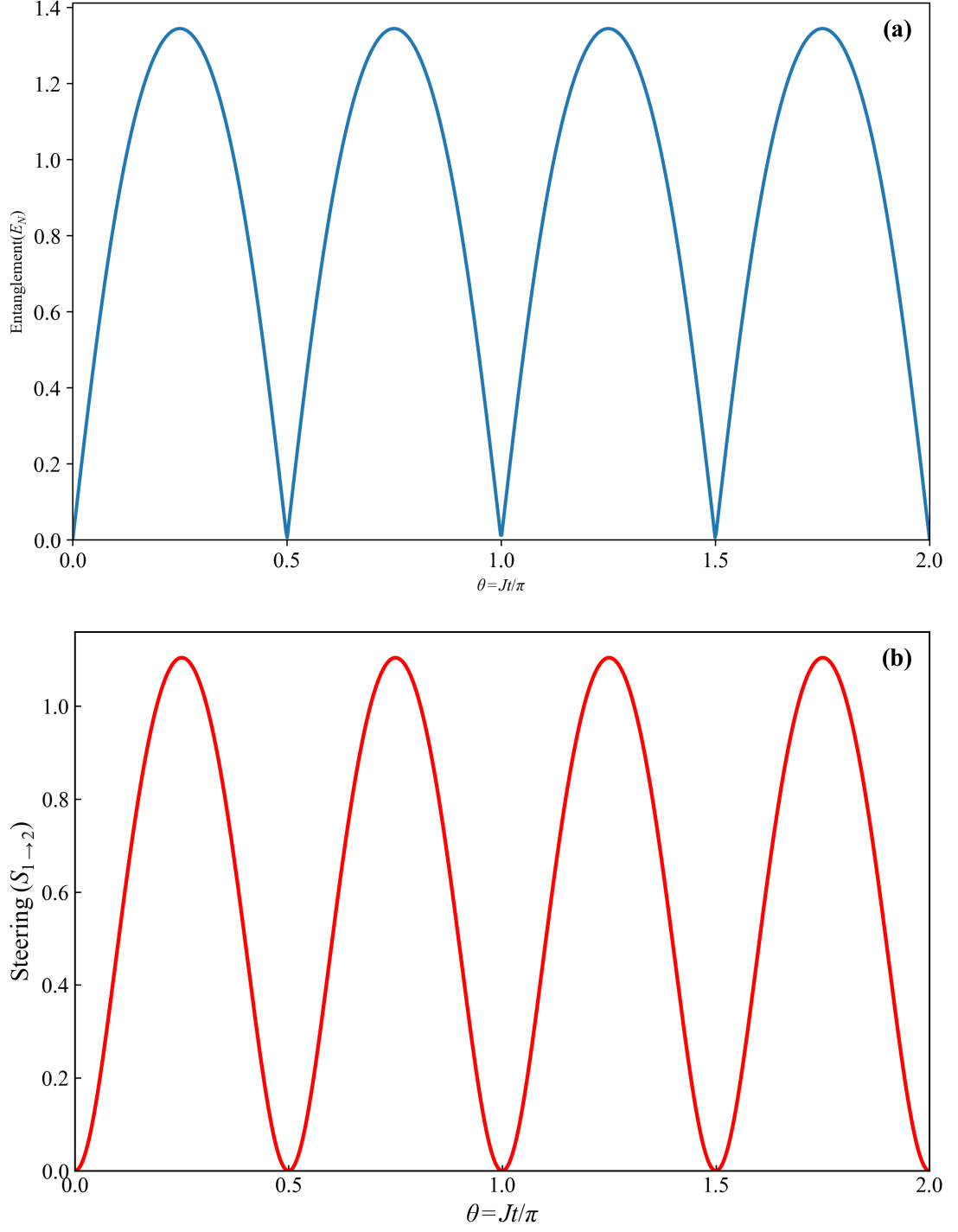


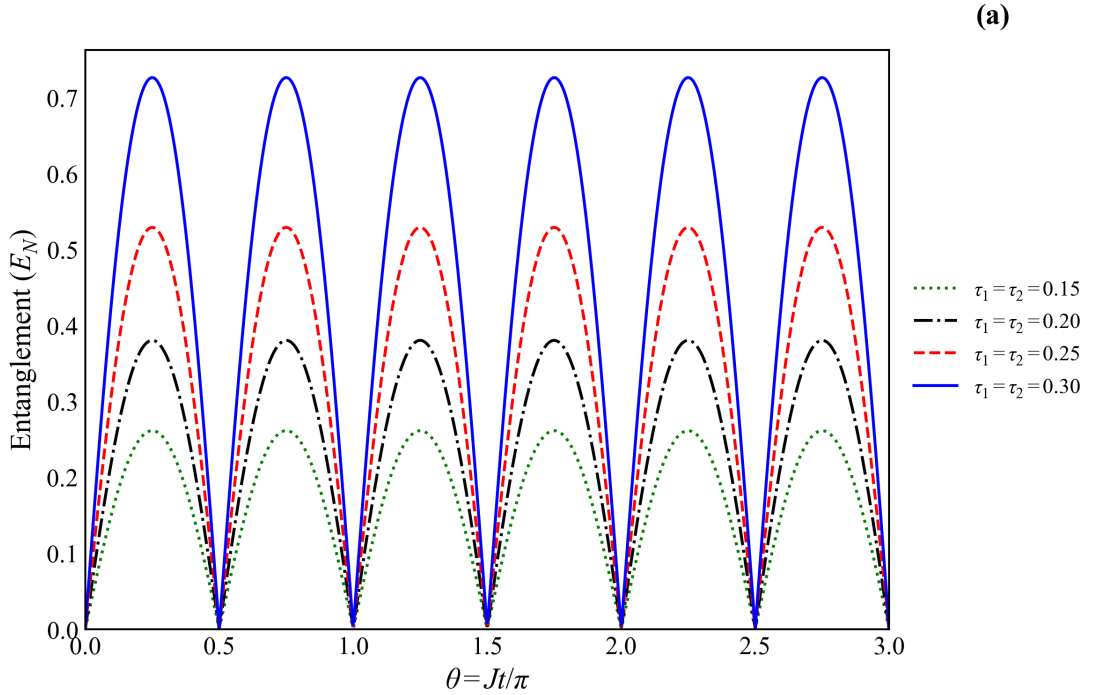
Fig4.1 shows two kind of quantum correlations: (a) Quantum Entanglement, (b) Quantum Steering against dimensionless time, for fixed parameters: $\mu_1 = \mu_2 = \mu = 1$, $\tau_1 = \tau_2 = \tau = 0.35$, $\kappa_1 = \kappa_2 = \kappa = 0$ and $\phi_1 = 0$ and $\phi_2 = \pi/2$.

4.2 Effect of Nonclassical Depth τ

In this section, we consider the effects of non classicality, referred to as nonclassicality depth, τ on Gaussian quantum steering and Gaussian Quantum entanglement, we have kept the purity μ and the loss parameter κ fixed in this analysis.

Figure 4.2 indicates that the greater the nonclassical depth the stronger becomes both Quantum Entanglement and Quantum Gaussian steering. The larger the τ the larger the peaks, the larger the Quantum correlations as they travel. This is to imply that the greater the nonclassicality of the initial Gaussian state is, the greater the ability it has to generate and maintain quantum correlations in the coupled waveguide system.

In addition, the Gaussian steering responds more to variation in the nonclassical depth in comparison to entanglement. This increased sensitivity is an indication that steering is more strict than entanglement requires stronger quadrature correlations.



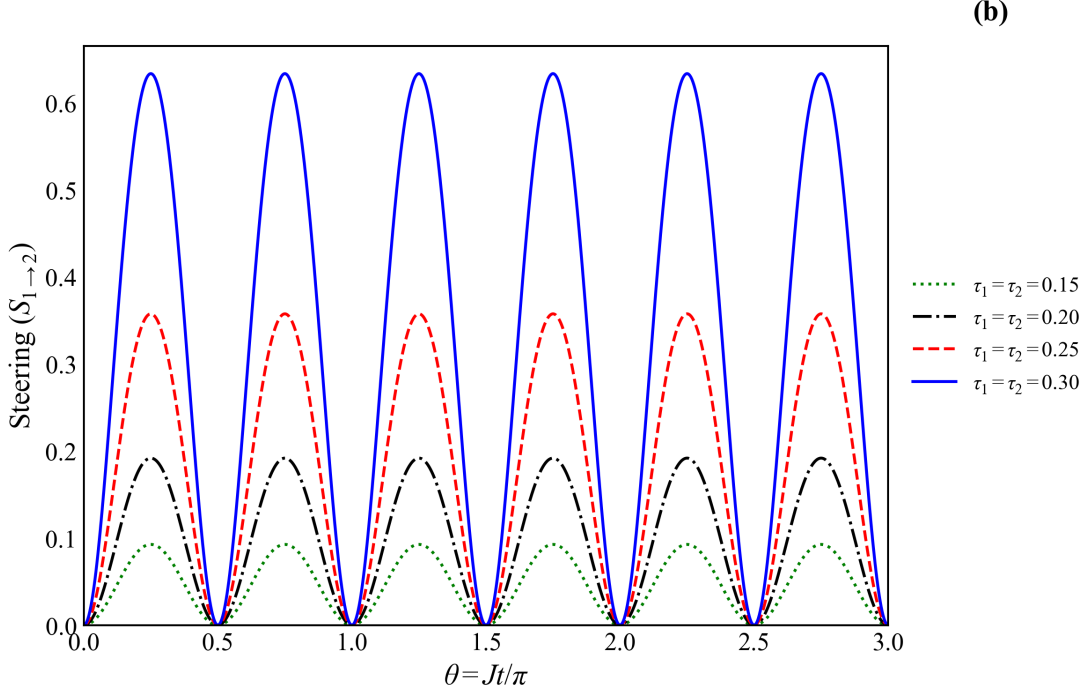


Fig4.2 represents two kinds of quantum correlations: (a) quantum entanglement and (b) quantum steering, with $\kappa_1 = \kappa_2 = \kappa = 0.05$, $\mu_1 = \mu_2 = \mu = 1$, $\phi_1 = 0$ and $\phi_2 = \pi/2$. In each case, the solid blue, dashed red, dash-dotted black, and dotted red lines correspond to $\tau_1 = \tau_2 = \tau = 0.15, 0.20, 0.25$, and 0.30 , respectively.

4.3 Effect of Purity μ

In this section, we are examining the impact of state purity μ on the development of Gaussian quantum steering and Gaussian Quantum Entanglement.

Fig. 4.3 demonstrates that in case of the reduction of the purity μ , the size of both quantum correlations decreases significantly.

We observe that, as the purity reduces, both Quantum steering and Quantum Entanglement decreases, indicating that these are vulnerable to noise and decoherence.

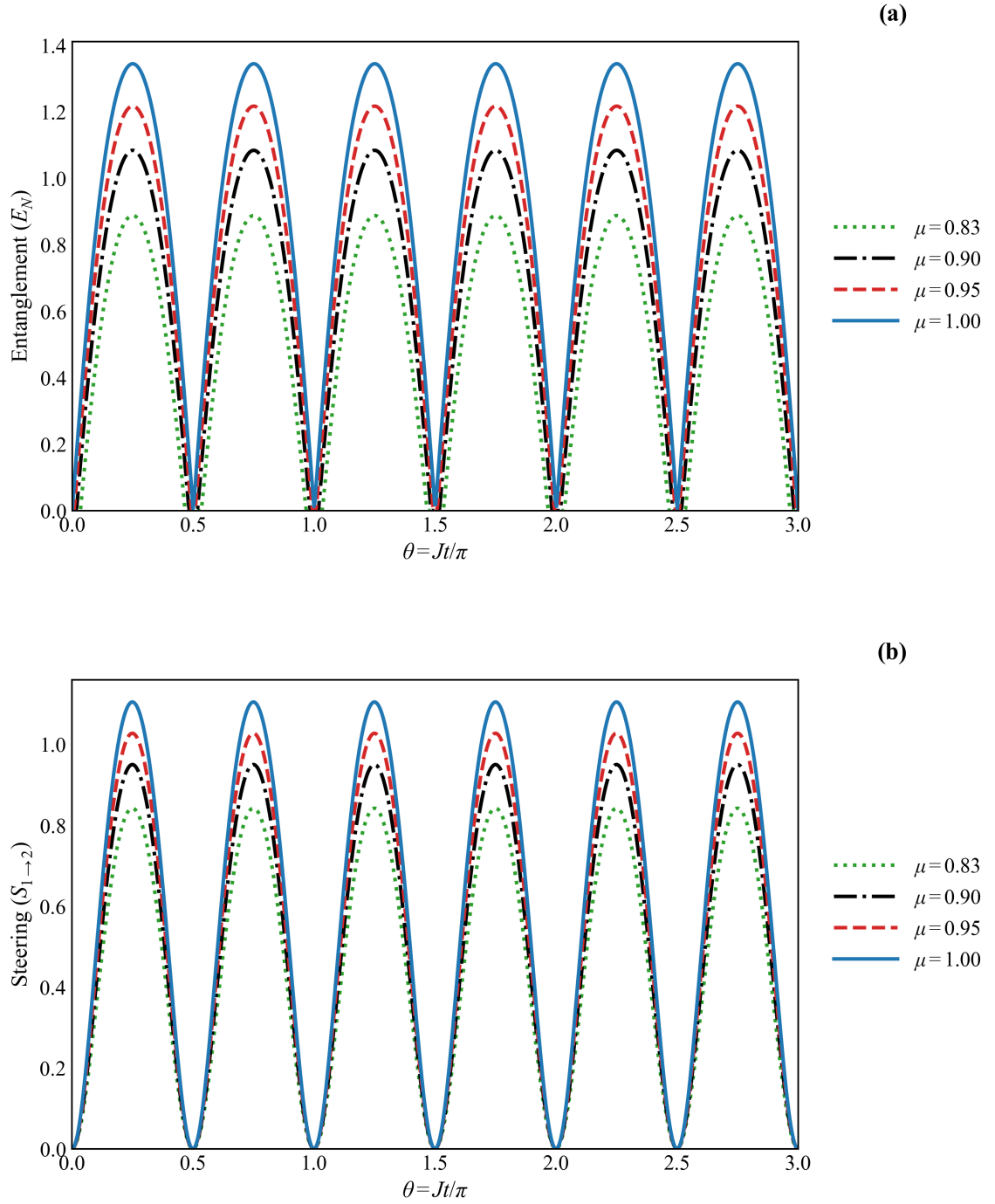
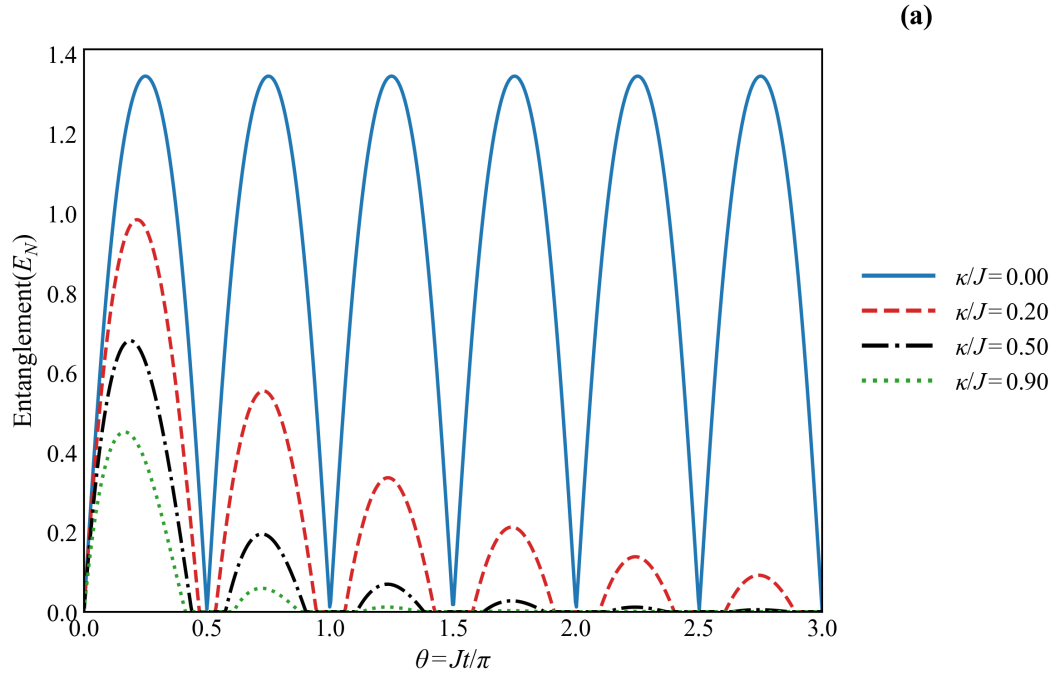


Fig4.2 shows two kinds of quantum correlations: (a) quantum entanglement and (b) quantum steering, with $\kappa_1 = \kappa_2 = \kappa = 0.05$, $\tau_1 = \tau_2 = \tau = 0.35$, $\phi_1 = 0$ and $\phi_2 = \pi/2$. In each case, the solid blue, dashed red, dashed-dotted black, and dotted red lines correspond to $\mu_1 = \mu_2 = \mu = 0.80, 0.90, 0.95$, and 1.0 , respectively.

4.4 Effect of Loss Parameter κ

Here, we can see what happens to Gaussian quantum steering and Gaussian entanglement in the coupled waveguide system when the loss parameter κ is changed.

Figure 4.4 indicates that the quantum correlations decrease significantly as the loss parameter increases. It is observed that Quantum Steering is less than Quantum Entanglement for all values of κ ob.



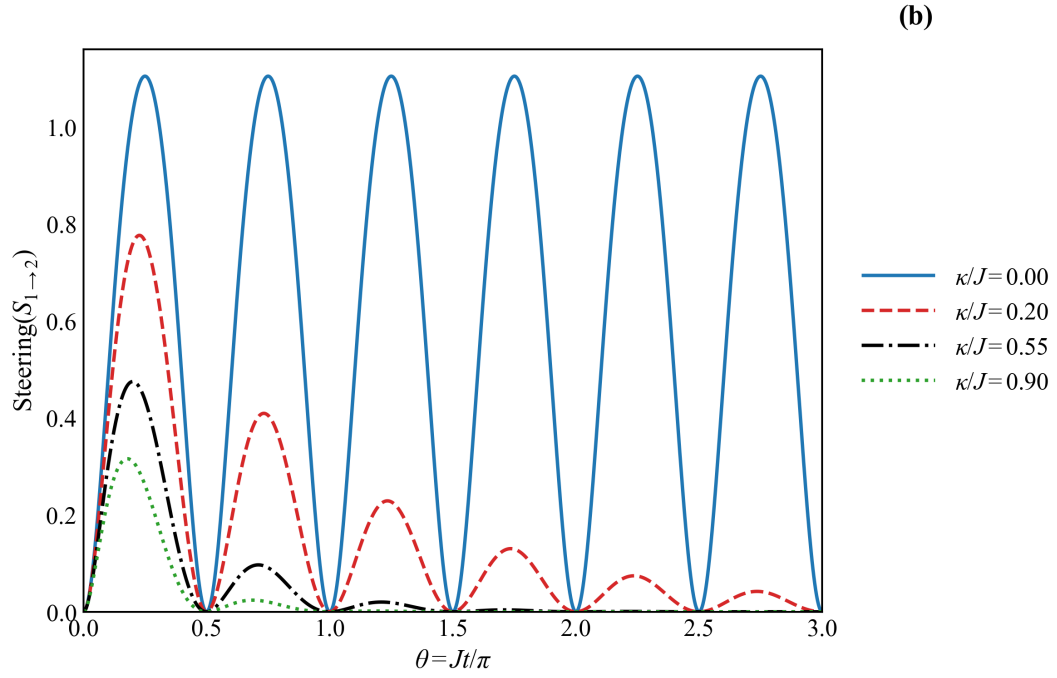


Fig4.3 represents two kinds of quantum correlations: (a) quantum entanglement and (b) quantum steering, with $\tau_1 = \tau_2 = \tau = 0.35$, $\phi_1 = 0$ and $\phi_2 = \pi/2$, and $\mu_1 = \mu_2 = \mu = 1$. In each case, the solid blue, dashed blue, dashed-dotted black, and dotted green lines correspond to $\kappa_1 = \kappa_2 = \kappa = 0, 0.20, 0.35$, and 0.60 , respectively.

4.5 Gaussian Steering and Entanglement in Squeezed States

here we explore the phenomenon of Gaussian quantum steering and Gaussian entanglement of a special type of Gaussian state, called squeezed state.

In the lossless regime, squeezing generates a strong Gaussian entanglement and steering which can be observed in Fig4.5. The strong quadrature correlations which squeezed states possess lead to the high peak values of both quantum correlations. where as loss effect entanglement and steering in squeezed states, quantum entanglement and quantum steering decreases with increasing loss rate. It is observed that for every value of κ quantum steering is less than quantum entanglement.

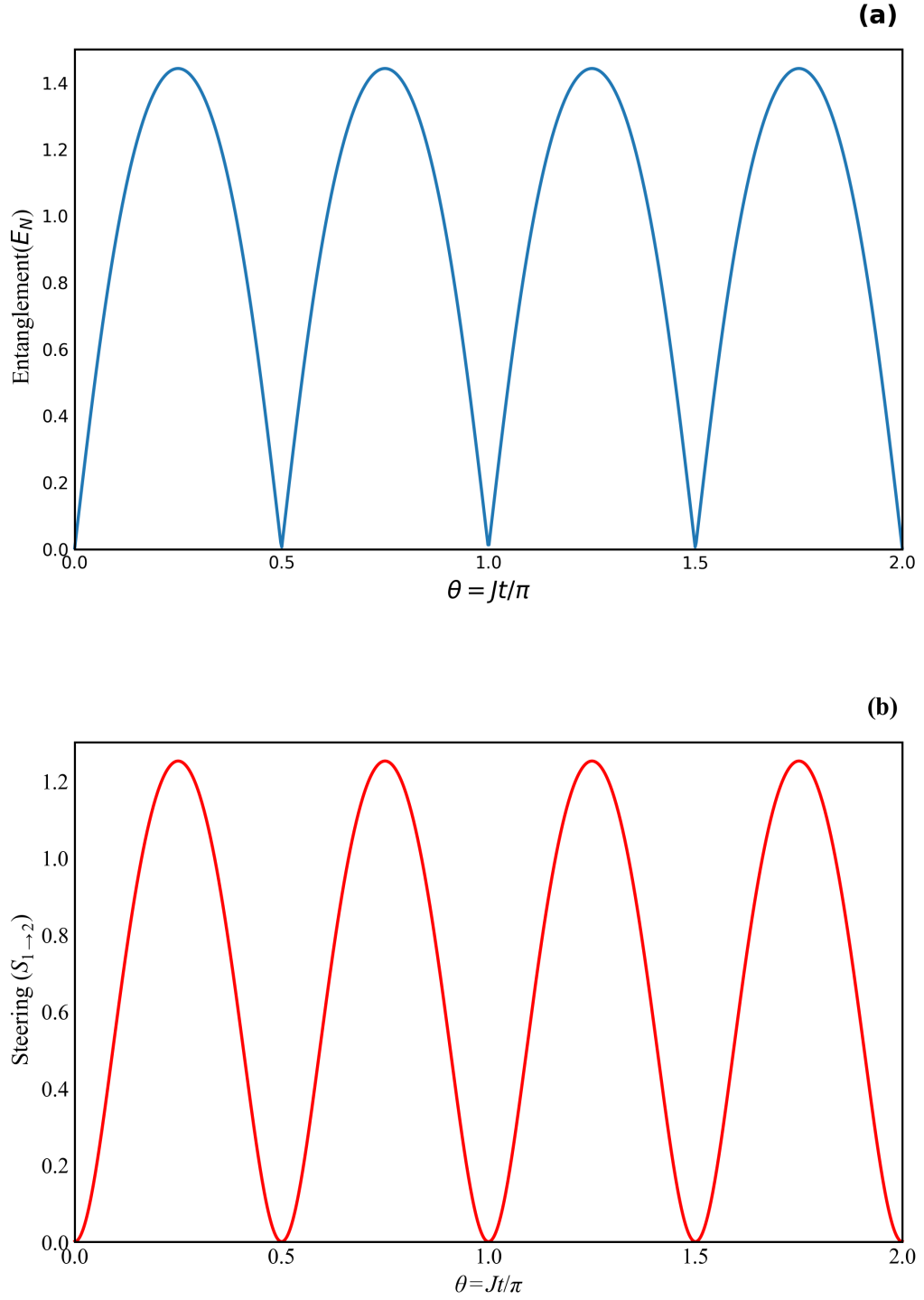
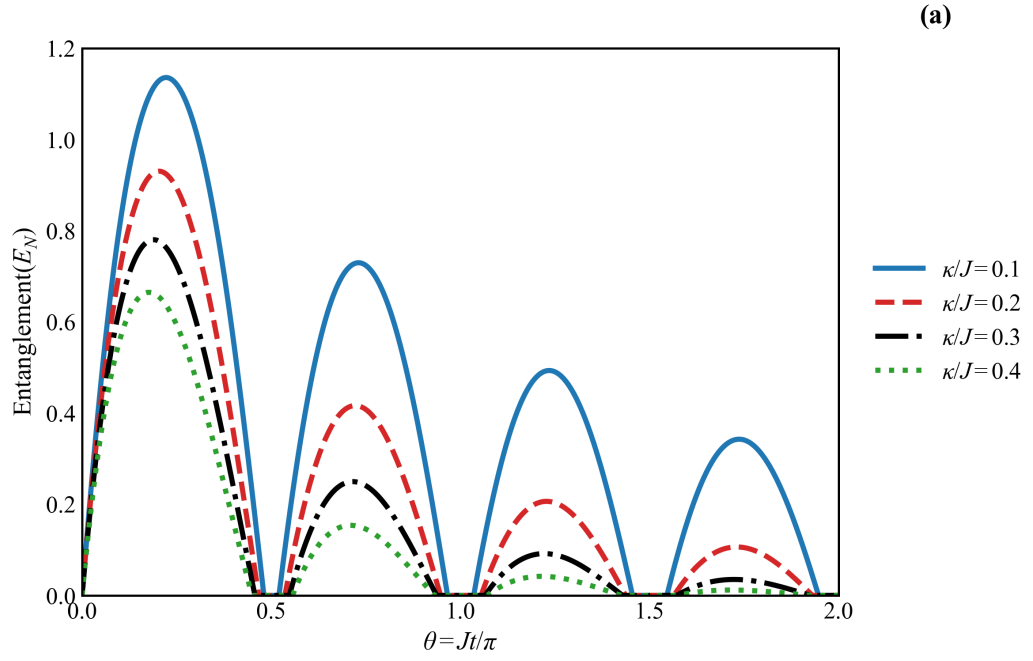


Fig4.3 shows two kinds of quantum correlations: (a) quantum entanglement and (b) quantum steering, with $\kappa_1 = \kappa_2 = \kappa = 0$ and squeezing factor $r = 0.5$. The solid blue curve represents entanglement, while the solid red curve represents steering under the same parameters.

4.5.1 Effect of loss on squeezed states

In this subsection we see the effect of loss on Quantum entanglement and Quantum steering in Squeezed states. In Fig4.6 we can see that both Entanglement and Steering is degraded by loss and the degradation depends on the magnitude of loss rate and steering is always less than entanglement for every value of κ considered.



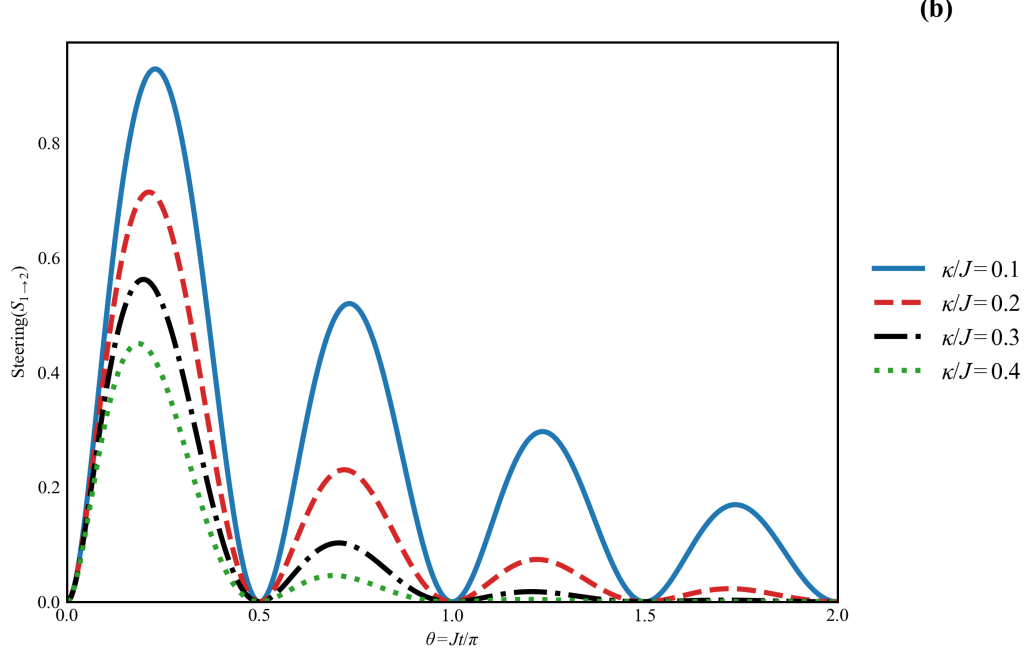


Fig4.4 shows two kinds of quantum correlations:(a) quantum entanglement and (b) quantum steering, with squeezing factor $r = 0.5$. In each case, the solid blue, dashed red, dash dotted black, and dotted green lines correspond to $\kappa_1 = \kappa_2 = \kappa = 0.1, 0.2, 0.3$, and 0.4 , respectively.

4.6 Effect of coupling strength

Finally, In this section we investigate the effect of the coupling strength J on Gaussian quantum entanglement and Gaussian steering, we kept the loss parameter κ fixed and set to unity in this analysis. The quantum correlations are plotted versus the scaled parameter κt , which represents the propagation time in the presence of loss. By fixing $\kappa=1$, we have observed variations in the dynamical behavior of Quantum entanglement and Quantum steering by changing the coupling strength J . As the coupling becomes stronger, both Gaussian entanglement and Gaussian steering amplitude maximum values are significantly attained. This demonstrates that a higher coupling assists in the generation and transport of quantum correlations across the waveguides that partially compensates the adverse impact of loss. Moreover, the stronger the coupling, the stronger the oscillations, which means that there is a bigger rate of coherent exchange of energy between the coupled modes. The findings indicate that better coupling enhances the generation of quantum correlations and both entanglement and steering peaks are

larger. The increase in the level of coupling facilitates easier movement of photons between the waveguides hence increasing the strength of the correlations between modes despite some level of loss. In the weaker connection, the correlations are narrow and fade away more rapidly with the wave transmission. The findings reveal that the enhancement of the formation of quantum correlations is achieved by stronger coupling, which provides higher peaks of both Gaussian entanglement and Gaussian steering. The increased coupling causes the photons to travel between the waveguides with lesser effort and hence the correlations between the modes become stronger even when a certain amount of loss is experienced. Conversely, when the coupling strengths are weaker the correlations generated are finite and they decay faster as they propagate. It is also observed that, regardless of the strength of the coupling between the two systems, Gaussian entanglement is always bigger than Gaussian steering.

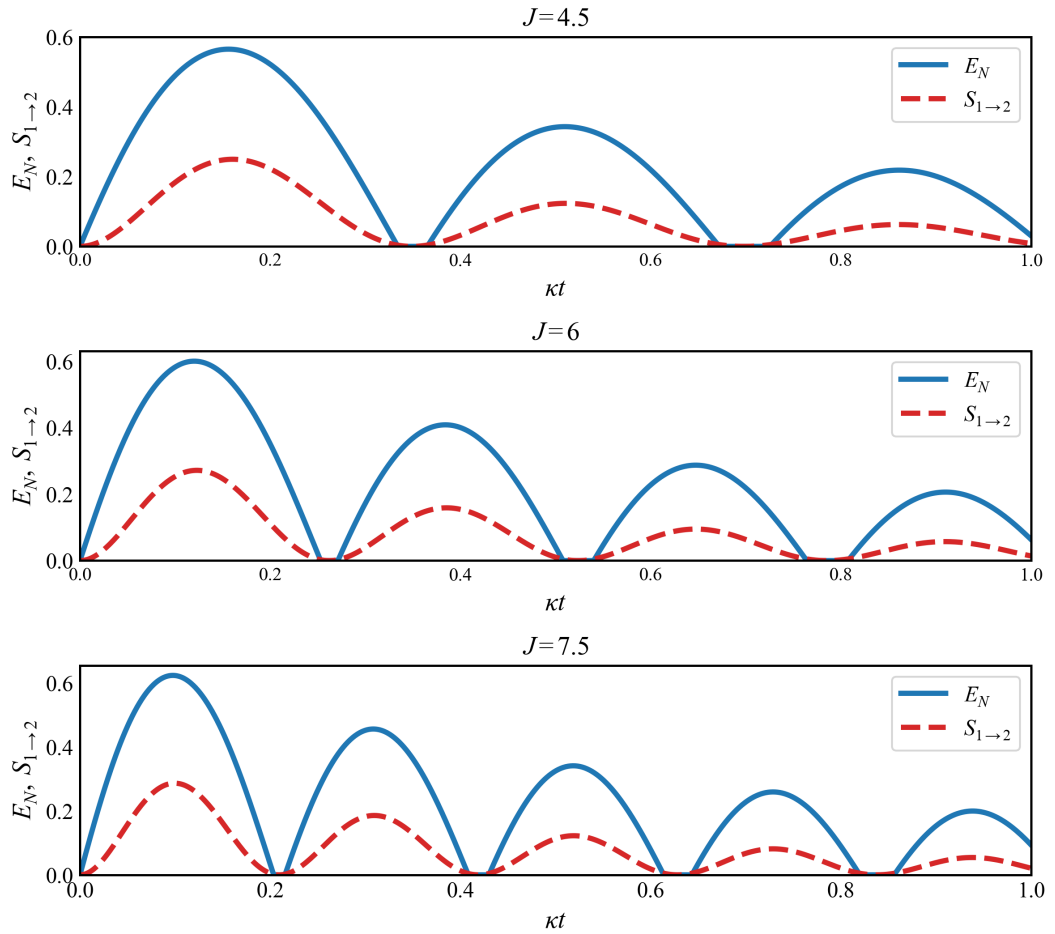


Fig4.5 Two kinds of quantum correlations: (a) quantum entanglement and (b) quan-

tum steering are plotted as functions of the dimensionless time κt for different coupling strengths $J = 4.5, 6$, and 7.5 . In each panel, the solid blue curve represents the entanglement E_N , while the dashed red curve represents the Gaussian steering $S_{1 \rightarrow 2}$.

4.7 Summary

The key findings of this chapter can be summarized as follows: This chapter has included an in-depth exploration of the dynamical performance of Gaussian quantum entanglement and Gaussian quantum steering within a coupled lossy waveguide system by the covariance matrix framework. This analysis has been done through the systematic investigation of the role of important physical parameters that determine the dynamics of the system. First of all we have learned the overall time evolution of entanglement and steering by setting the nonclassical depth, purity and loss parameters fixed. The experiment found an oscillatory behavior of both quantum correlations as functions of the normalized propagation time, and was due to the coherent mode-coupling interaction between the waveguides. It was found that maxima and minima in entanglement as well as steering take identical propagation lengths which validated their physical origins. As the process continued, the Gaussian steering was always smaller than the Gaussian entanglement, as it was known in the hierarchy of quantum correlations. After that, the influence of nonclassical depth was analyzed. It was revealed that the increase in nonclassical depth positively affected the Gaussian entanglement and steering, and steering was more sensitive to nonclassicality changes. This shows the more high demands of steering, which have high quadrature correlation requirements compared to entanglement. The effect of purity by the state was then examined. The purity decrease resulted in a major degradation of the entanglement and steering, thus showing the susceptibility of the two to noise and decoherence. This establishes that the only way to maintain quantum correlations in real systems is to ensure that the systems are higher purified. The effects of the loss was also examined. An increment in the loss parameter led to a sharp reduction of the quantum correlations i.e quantum entanglement and quantum steering. We then studied the role of squeezing with reference to squeezed Gaussian states. Squeezing in the absence of its loss has high entanglement and steering due to the increased quadrature correlations. Both of the correlations decreases by introducing

loss. Although loss decreases quantum correlations but it does not vanish them, they are correlated even with effect of loss. Lastly, the influence of the strength of coupling was investigated by holding the loss parameter constant. It was discovered that high coupling increases evanescent coupling signifying more efficient transfer of photons, hence increases number of oscillation which produce great improvement in the amplitudes and persistence of entanglement and steering, which even compensate the effect of loss parameter which is kept 1. However, steering was always less than entanglement which proves the hierarchy. On the whole, this chapter confirms that the nonclassicality, purity, loss, squeezing, and the strength of the coupling in coupled waveguide systems have a strong impact on the Gaussian quantum entanglement and steering. Both entanglement and steering are found to be robust and steering is less than entanglement for all values which proves the quantum hierarchy.

Chapter 5

Conclusion

In this thesis, the dynamical behaviour of Gaussian Quantum Steering and Gaussian Quantum Entanglement have been investigated in a coupled lossy waveguide system using the covariance matrix formalism within the framework of continuous variable quantum optics. The reason behind the study is that it is important to get the insights on the impact of realistic dissipation and system parameters on the persistence and utility of quantum correlations in integrated photonic platforms.

The analysis has revealed that, quantum entanglement and quantum steering have oscillatory behaviour with respect to propagation time which is explained by coherent transfer of energy between the coupled waveguide modes. These oscillations are indicative of the underlying evanescent mechanism of coupling and is a conclusive evidence of interaction between coherent modes within the system. However, the loss of correlations between the two in the presence of optical loss is what causes a gradual decay of both of the correlations, which in turn demonstrates the inevitable effect of dissipation in realistic photonic devices.

One of the key findings of the given work is that the hierarchical character of quantum correlations indeed exists: quantum steering is seen to be more vulnerable than quantum entanglement in the same physical conditions and is less in magnitude than entanglement. This behaviour is seen in all regimes of parameters studied and in total accord with the theoretical hierarchy between Quantum Entanglement and Quantum Steering. Systematic study of the quantum correlation with significant physical parameters was done. The increase of the nonclassical depth and purity is discovered to increase entanglement and steering, and thus increase their resistance to loss. On the other hand, an increased loss coefficient increases the rate of quantum correlation degradation. The coupling parameter varies the rate of exchange of quantum correlations between the modes, it compensates the effect of loss parameter which is kept as 1 for this special case only.

The study of squeezed Gaussian states goes further to indicate that, whereas ideal lossless systems can sustain strong quantum correlations, realistic loss, although decreases quantum correlations but entanglement and steering are still robust even in the presence of loss.

To sum up, this paper has proved that useful amounts of Gaussian quantum correlations can be maintained in integrated waveguide systems with dissipation that cannot be avoided, offering important information about their practical constraints and abilities. The results provide a design and optimization rule of the photonic quantum devices, and in instances where asymmetric quantum resources are required, including but not limited to one-sided device-independent quantum key distribution, where quantum steering is key.

Chapter 6

Conclusions

Moment Equations for the Equal-Loss Two-Mode Coupler

We consider two bosonic modes with annihilation operators $\hat{a}_1$ and $\hat{a}_2$, coupled through the passive beam-splitter Hamiltonian

$$\hat{H} = J(\hat{a}_1^\dagger \hat{a}_2 + \hat{a}_2^\dagger \hat{a}_1),$$

and subject to equal damping rates

$$\kappa_1 = \kappa_2 = \kappa.$$

The relevant complex Gaussian moments are defined as

$$\begin{aligned} n_1 &= \langle \hat{a}_1^\dagger \hat{a}_1 \rangle, & n_2 &= \langle \hat{a}_2^\dagger \hat{a}_2 \rangle, \\ m_1 &= \langle \hat{a}_1 \hat{a}_1 \rangle, & m_2 &= \langle \hat{a}_2 \hat{a}_2 \rangle, \\ s &= \langle \hat{a}_1 \hat{a}_2 \rangle, & c &= \langle \hat{a}_1^\dagger \hat{a}_2 \rangle. \end{aligned}$$

Using the Heisenberg–Langevin equations (noise averages vanish for a vacuum bath), one obtains the coupled first-order differential equations

$$\begin{aligned} \dot{n}_1 &= -\kappa n_1 + iJ(c^* - c), \\ \dot{n}_2 &= -\kappa n_2 + iJ(c - c^*), \\ \dot{m}_1 &= -\kappa m_1 - 2iJ s, \\ \dot{m}_2 &= -\kappa m_2 - 2iJ s, \\ \dot{s} &= -\kappa s - iJ(m_1 + m_2), \\ \dot{c} &= -\kappa c + iJ(n_2 - n_1). \end{aligned}$$

The initial conditions used throughout this work are

$$\begin{aligned} n_1(0) &= n_{10}, & n_2(0) &= n_{20}, \\ m_1(0) &= m_{10}, & m_2(0) &= m_{20}, \\ s(0) &= 0, & c(0) &= 0, \end{aligned}$$

corresponding to uncorrelated input Gaussian states.

Analytic Solution Under Equal Loss

The system above admits a closed-form solution when $\kappa_1 = \kappa_2 = \kappa$. Define the constants

$$\begin{aligned} C_1 &= n_1(0) + \frac{1}{2}, & C_2 &= n_2(0) + \frac{1}{2}, \\ D_1 &= m_1(0), & D_2 &= m_2(0). \end{aligned}$$

Then the exact solutions for all Gaussian moments are

$$n_1(t) = \frac{e^{-\kappa t}}{2} \left[(C_1 + C_2 - 1) + (C_1 - C_2) \cos(2Jt) \right],$$

$$n_2(t) = \frac{e^{-\kappa t}}{2} \left[(C_1 + C_2 - 1) - (C_1 - C_2) \cos(2Jt) \right],$$

$$c(t) = +\frac{i e^{-\kappa t}}{2} (C_2 - C_1) \sin(2Jt),$$

$$m_1(t) = \frac{e^{-\kappa t}}{2} \left[(D_1 + D_2) \cos(2Jt) + (D_1 - D_2) \right],$$

$$m_2(t) = \frac{e^{-\kappa t}}{2} \left[(D_1 + D_2) \cos(2Jt) + (D_2 - D_1) \right],$$

$$s(t) = -\frac{i e^{-\kappa t}}{2} (D_1 + D_2) \sin(2Jt).$$

These expressions are used directly in the numerical evaluations of Gaussian steering and entanglement throughout this thesis. what is issue with this

Chapter 7

Model System and Theoretical Formulations

7.1 physical system configuration

our system consist of two identical single mode waveguides which are placed parallel to each other in a close distance such that their evanescent feild overlap and enable coupling between them. The optical modes in each waveguide are represented by creation and anihilation operator a_1 and a_2 where a_1 represents optical mode in waveguide a_1 and a_2 represents optical mode in waveguide a_2 . Each waveguide mode act as a bosonic harmonic oscillator.

7.1.1 system Hamiltonian

The hamiltonian for our system is given by

$$H = \hbar\omega \left(a_1^\dagger a_1 + a_2^\dagger a_2 \right) + \hbar J \left(a_1^\dagger a_2 + a_2^\dagger a_1 \right). \quad (7.1)$$

Here, a_1 and a_2 are the bosonic field operators for the optical modes in each waveguide, satisfying the commutation relations $[a_1, a_1^\dagger] = 1$ and $[a_2, a_2^\dagger] = 1$.

the symbol ω is angular frequency of the each waveguide mode and the parameter J in the second part of the hamiltonian represents the coupling strength between the two waveguides and its value depends on the seperation distance between the waveguides.

The hamiltonian consists of two main parts:

Free energy terms: correspond to the free energy of the individual waveguide modes,

$$\hbar\omega \left(a_1^\dagger a_1 + a_2^\dagger a_2 \right).$$

Coupling terms: account for the evanescent coupling between the two waveguide modes,

$$\hbar J \left(a_1^\dagger a_2 + a_2^\dagger a_1 \right).$$

This term is responsible for the transfer of energy between the two waveguide modes and is the main focus of interest in this work.

we can determine how this system evolve with loss by quantum louville equation.

$$\dot{\rho} = -\frac{i}{\hbar} [H, \rho] + \mathcal{L}\rho \quad (7.2)$$

this equation is known as quantum Liouville equation. It has two parts:

The Commutator:

$$[H, \rho]$$

which generates time evolution and is responsible for oscillatory exchange of energy and quantum correlation between the waveguides due to J .

Lindblad($L\rho$) term : deals with material loss, decoherence and decays due to interaction with the environment

$$\begin{aligned} \mathcal{L}\rho = & -k_1 \left(a_1^\dagger a_1 \rho + \rho a_1^\dagger a_1 - 2a_1 \rho a_1^\dagger \right) \\ & - k_2 \left(a_2^\dagger a_2 \rho + \rho a_2^\dagger a_2 - 2a_2 \rho a_2^\dagger \right), \end{aligned} \quad (7.3)$$

k_1 and k_2 represent the loss rates of modes a and b , and $a, a^\dagger$ are the annihilation and creation operators for mode a , while ρ is the density matrix. b and $b^\dagger$ are vice versa for mode b .

7.2 Dynamical equations of the model system

The time evolution of any system observable O is obtained from

$$\frac{d}{dt}\langle O \rangle = \text{Tr}(O\dot{\rho}) = -i\langle [O, H] \rangle + \langle \mathcal{L}^\dagger(O) \rangle. \quad (7.4)$$

7.2.1 Equations of motion for the moments

Using

$$\frac{d}{dt}\langle O \rangle = \text{Tr}(O\dot{\rho}),$$

Equation number(3.2), the coupled equations of motion becomes:

$$\frac{d}{dt}\langle a_1^\dagger a_1 \rangle = iJ\left(\langle a_2^\dagger a_1 \rangle - \langle a_1^\dagger a_2 \rangle\right) - k_1\langle a_1^\dagger a_1 \rangle, \quad (7.5)$$

$$\frac{d}{dt}\langle a_2^\dagger a_2 \rangle = iJ\left(\langle a_1^\dagger a_2 \rangle - \langle a_2^\dagger a_1 \rangle\right) - k_2\langle a_2^\dagger a_2 \rangle, \quad (7.6)$$

$$\frac{d}{dt}\langle a_1^\dagger a_2 \rangle = iJ\left(\langle a_2^\dagger a_2 \rangle - \langle a_1^\dagger a_1 \rangle\right) - \frac{k_1 + k_2}{2}\langle a_1^\dagger a_2 \rangle, \quad (7.7)$$

$$\frac{d}{dt}\langle a_2^\dagger a_1 \rangle = iJ\left(\langle a_1^\dagger a_1 \rangle - \langle a_2^\dagger a_2 \rangle\right) - \frac{k_1 + k_2}{2}\langle a_2^\dagger a_1 \rangle, \quad (7.8)$$

$$\frac{d}{dt}\langle a_1 a_1 \rangle = -2iJ\langle a_1 a_2 \rangle - k_1\langle a_1 a_1 \rangle, \quad (7.9)$$

$$\frac{d}{dt}\langle a_2 a_2 \rangle = -2iJ\langle a_1 a_2 \rangle - k_2\langle a_2 a_2 \rangle, \quad (7.10)$$

$$\frac{d}{dt}\langle a_1 a_2 \rangle = -iJ\left(\langle a_1 a_1 \rangle + \langle a_2 a_2 \rangle\right) - \frac{k_1 + k_2}{2}\langle a_1 a_2 \rangle. \quad (7.11)$$

so we have got seven differential equations, now we are moving towards the solutions of these equations

7.2.2 analytical solutions of the moment equations

For identical waveguides, we set

$$k_1 = k_2 = \kappa. \quad (7.12)$$

Without imposing any restriction on the initial correlations, the seven second-order moments admit the following closed-form solutions:

$$\begin{aligned} \langle a_1^\dagger a_1 \rangle(t) = & \frac{e^{-\kappa t}}{2} \left[\langle a_1^\dagger a_1 \rangle_0 + \langle a_2^\dagger a_2 \rangle_0 + (\langle a_1^\dagger a_1 \rangle_0 - \langle a_2^\dagger a_2 \rangle_0) \cos(2Jt) \right. \\ & \left. + i(\langle a_2^\dagger a_1 \rangle_0 - \langle a_1^\dagger a_2 \rangle_0) \sin(2Jt) \right], \end{aligned} \quad (7.13)$$

$$\begin{aligned} \langle a_2^\dagger a_2 \rangle(t) = & \frac{e^{-\kappa t}}{2} \left[\langle a_1^\dagger a_1 \rangle_0 + \langle a_2^\dagger a_2 \rangle_0 - (\langle a_1^\dagger a_1 \rangle_0 - \langle a_2^\dagger a_2 \rangle_0) \cos(2Jt) \right. \\ & \left. - i(\langle a_2^\dagger a_1 \rangle_0 - \langle a_1^\dagger a_2 \rangle_0) \sin(2Jt) \right], \end{aligned} \quad (7.14)$$

$$\langle a_1^\dagger a_2 \rangle(t) = e^{-\kappa t} \left[\langle a_1^\dagger a_2 \rangle_0 \cos(2Jt) - \frac{i}{2} (\langle a_1^\dagger a_1 \rangle_0 - \langle a_2^\dagger a_2 \rangle_0) \sin(2Jt) \right], \quad (7.15)$$

$$\langle a_2^\dagger a_1 \rangle(t) = e^{-\kappa t} \left[\langle a_2^\dagger a_1 \rangle_0 \cos(2Jt) + \frac{i}{2} (\langle a_1^\dagger a_1 \rangle_0 - \langle a_2^\dagger a_2 \rangle_0) \sin(2Jt) \right], \quad (7.16)$$

$$\begin{aligned} \langle a_1 a_1 \rangle(t) = & \frac{e^{-\kappa t}}{2} \left[\langle a_1 a_1 \rangle_0 - \langle a_2 a_2 \rangle_0 + (\langle a_1 a_1 \rangle_0 + \langle a_2 a_2 \rangle_0) \cos(2Jt) \right. \\ & \left. - 2i \langle a_1 a_2 \rangle_0 \sin(2Jt) \right], \end{aligned} \quad (7.17)$$

$$\begin{aligned} \langle a_2 a_2 \rangle(t) = & \frac{e^{-\kappa t}}{2} \left[-(\langle a_1 a_1 \rangle_0 - \langle a_2 a_2 \rangle_0) + (\langle a_1 a_1 \rangle_0 + \langle a_2 a_2 \rangle_0) \cos(2Jt) \right. \\ & \left. - 2i \langle a_1 a_2 \rangle_0 \sin(2Jt) \right], \end{aligned} \quad (7.18)$$

$$\langle a_1 a_2 \rangle(t) = e^{-\kappa t} \left[\langle a_1 a_2 \rangle_0 \cos(2Jt) - \frac{i}{2} (\langle a_1 a_1 \rangle_0 + \langle a_2 a_2 \rangle_0) \sin(2Jt) \right]. \quad (7.19)$$

Although seven coupled moment equations are obtained, they do not all correspond to independent dynamical variables. In particular, the inter-mode coherence terms $\langle a_1^\dagger a_2 \rangle$ and $\langle a_2^\dagger a_1 \rangle$ are related by Hermitian conjugation, i.e., $\langle a_2^\dagger a_1 \rangle = \langle a_1^\dagger a_2 \rangle^*$. Therefore, only six independent moments are sufficient to fully describe the system dynamics, while the remaining coherence follows directly from this relation.

Define the constants

$$C_1 = \langle a_1^\dagger a_1 \rangle_0 + \frac{1}{2}, \quad C_2 = \langle a_2^\dagger a_2 \rangle_0 + \frac{1}{2},$$

$$D_1 = \langle a_1 a_1 \rangle_0, \quad D_2 = \langle a_2 a_2 \rangle_0.$$

Then the exact solutions for all Gaussian moments are

$$n_1(t) = \frac{e^{-\kappa t}}{2} \left[(C_1 + C_2 - 1) + (C_1 - C_2) \cos(2Jt) \right],$$

$$n_2(t) = \frac{e^{-\kappa t}}{2} \left[(C_1 + C_2 - 1) - (C_1 - C_2) \cos(2Jt) \right],$$

$$c(t) = +\frac{i e^{-\kappa t}}{2} (C_2 - C_1) \sin(2Jt),$$

$$m_1(t) = \frac{e^{-\kappa t}}{2} \left[(D_1 + D_2) \cos(2Jt) + (D_1 - D_2) \right],$$

$$m_2(t) = \frac{e^{-\kappa t}}{2} \left[(D_1 + D_2) \cos(2Jt) + (D_2 - D_1) \right],$$

$$s(t) = -\frac{i e^{-\kappa t}}{2} (D_1 + D_2) \sin(2Jt).$$

where, For convenience, we introduce the following time-dependent second-order moments:

$$n_1(t) = \langle a_1^\dagger a_1 \rangle(t), \quad n_2(t) = \langle a_2^\dagger a_2 \rangle(t),$$

$$c(t) = \langle a_1^\dagger a_2 \rangle(t), \quad m_1(t) = \langle a_1 a_1 \rangle(t), \quad m_2(t) = \langle a_2 a_2 \rangle(t), \quad s(t) = \langle a_1 a_2 \rangle(t).$$

These expressions are used directly in the numerical evaluations of Gaussian steering and entanglement throughout this thesis. what is issue with this

For independently prepared input modes (no initial inter-mode correlations), $\langle a_1^\dagger a_2 \rangle_0 = \langle a_1 a_2 \rangle_0 = 0$, and the corresponding terms drop out, yielding the simplified expressions used later in this thesis. ‘

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